ROSMASTER-M1PRO User Manual

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0. Preparation

- Refer to [Robot Configuration and Operation Guide — Assembly Video] to complete the assembly of the car's components and power on the car.

1. First-Time Use

1.1 Entering the Configuration Wizard

- After the car finishes booting, the OLED screen displays the car's hotspot `IP:192.168.1.11`



- Connect your phone or computer to the car's hotspot, open a browser, and access the initial configuration wizard interface:

```
http://192.168.1.11:8080
```

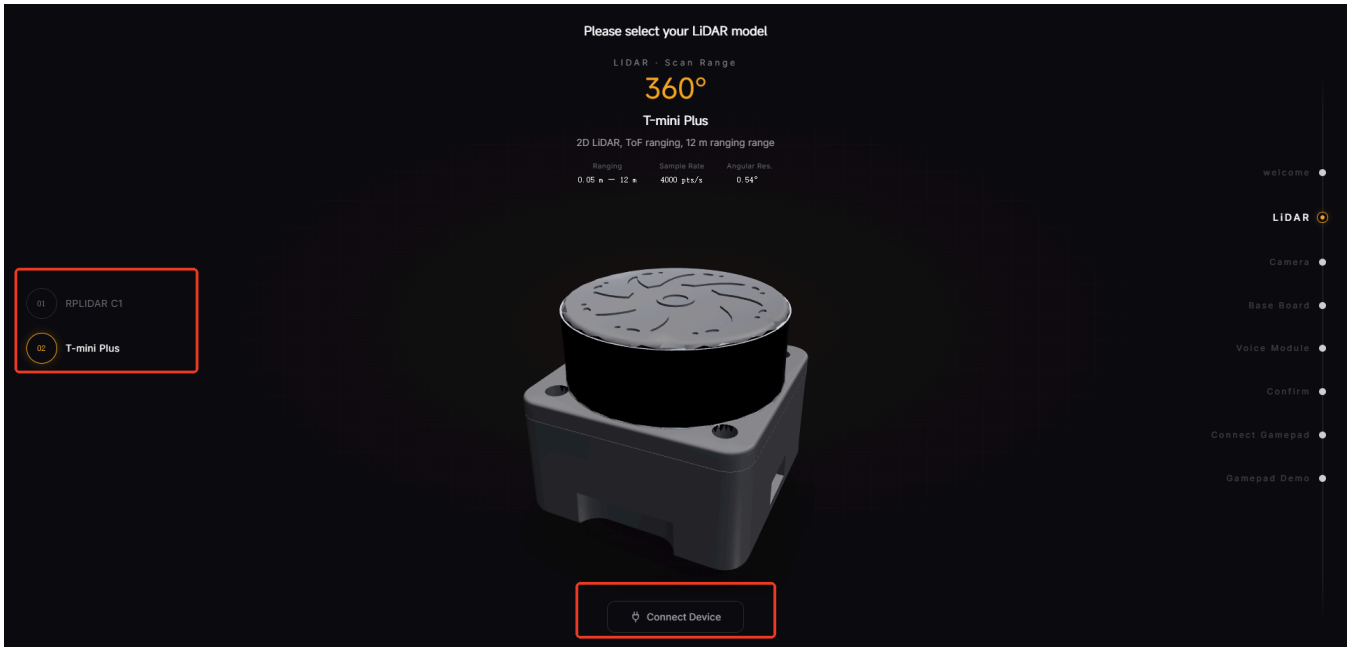
- You can switch the display language (Chinese/English) at the bottom of the interface



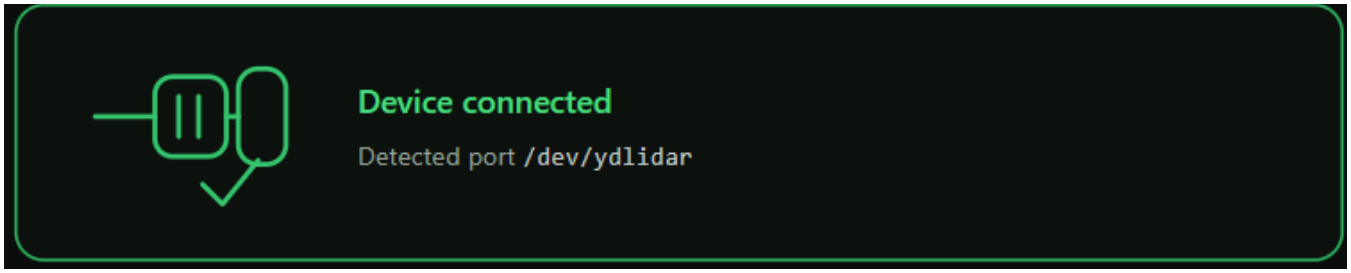
1.2 Selecting Hardware Configuration

- ROSMASTER-M1PRO provides different cameras [depth camera, USB monocular camera] and LiDARs [Slamtec C1, Tmini Plus] to choose from. Select your camera and LiDAR version in the configuration wizard.

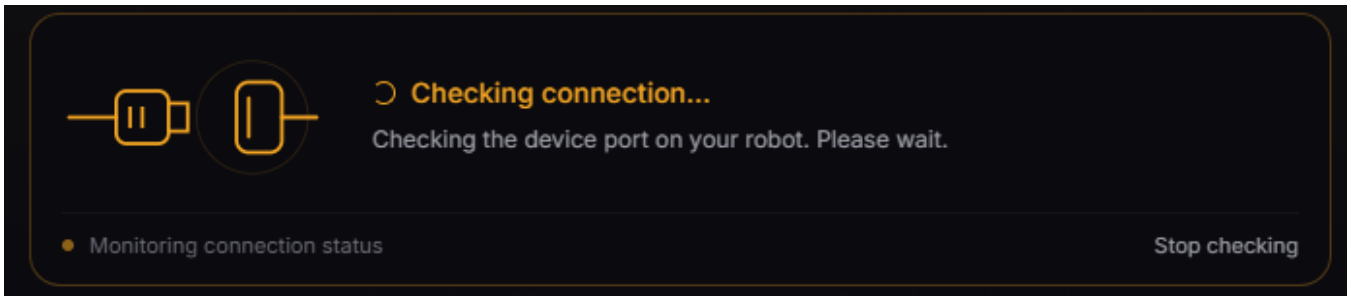
- LiDAR selection: select your LiDAR model on the left, then click Connect Device



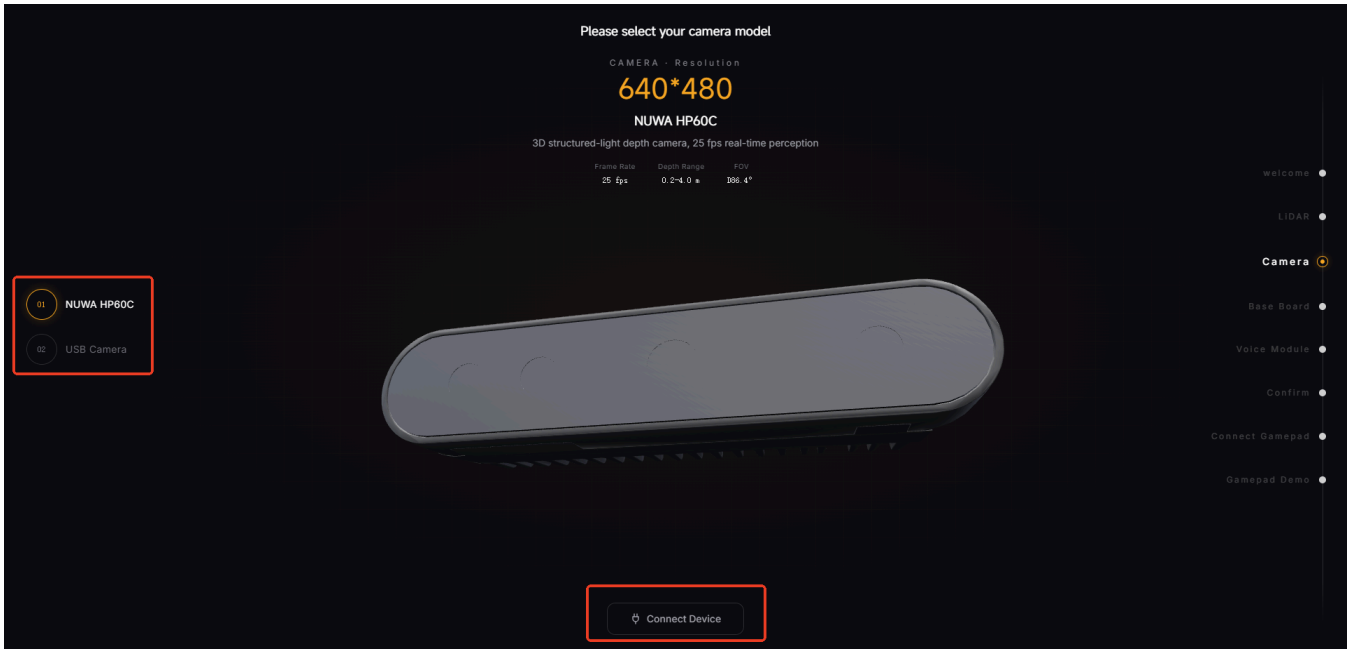
- If the device is wired correctly, a message indicating a successful device connection will appear



- If a message indicating it is waiting for the device to connect appears, check whether the device's wiring is loose, or unplug and replug the device

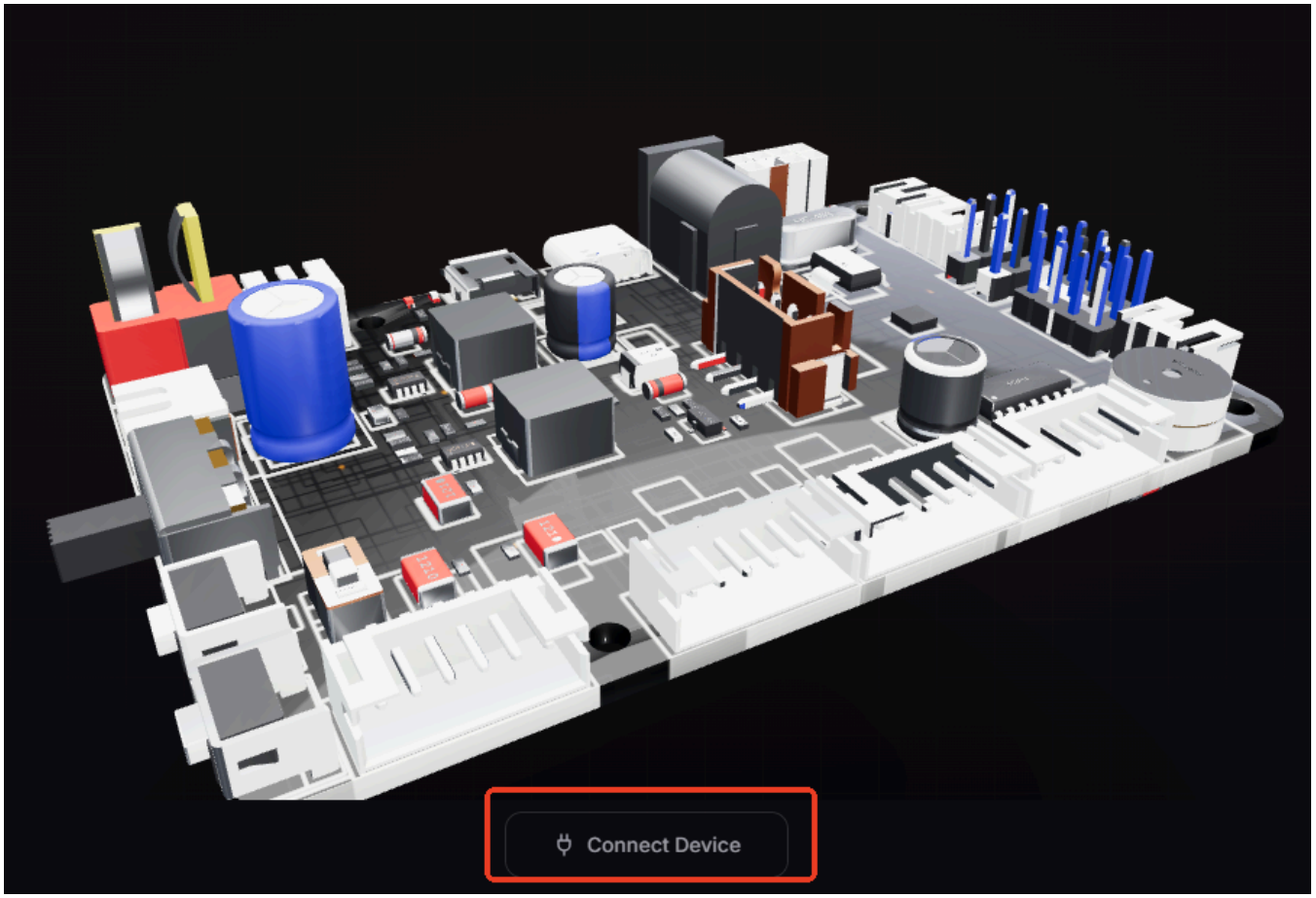


- Camera selection: perform the same operation—select the device on the left, then click Connect Device

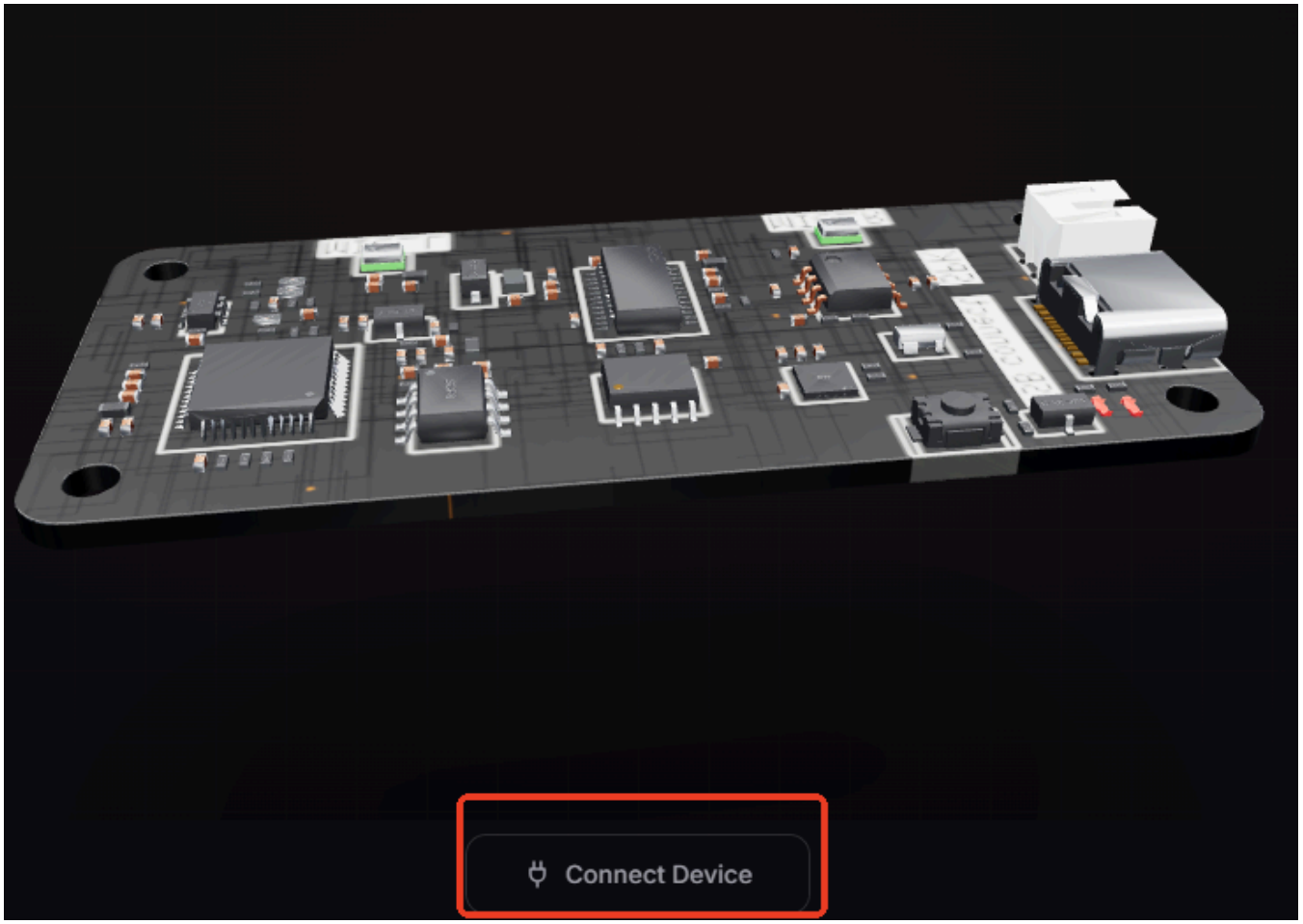


> **Note**
>
> The other device pages work the same way. The remaining devices do not require model selection; simply connect the device to determine whether the device wiring is properly connected during assembly

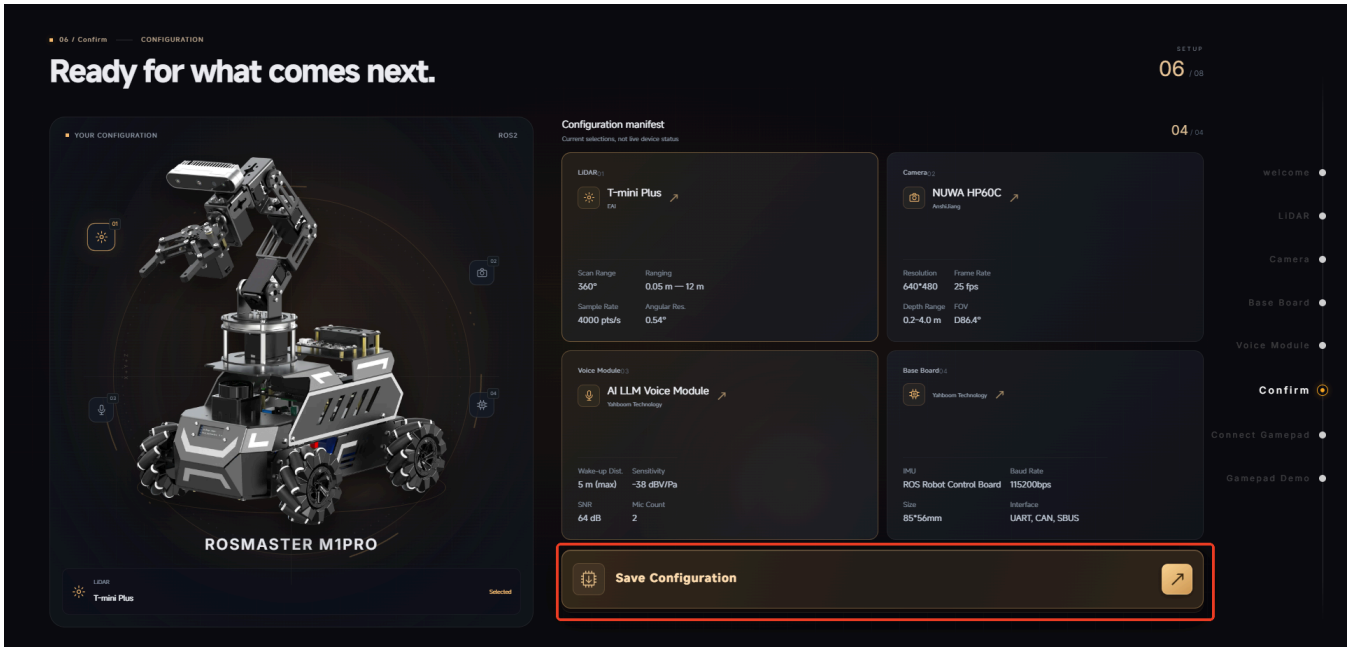
- Robot control board: click Connect Device



- Voice module: click Connect Device



- After completing camera and LiDAR selection and confirming that the devices are wired correctly, click Save Configuration

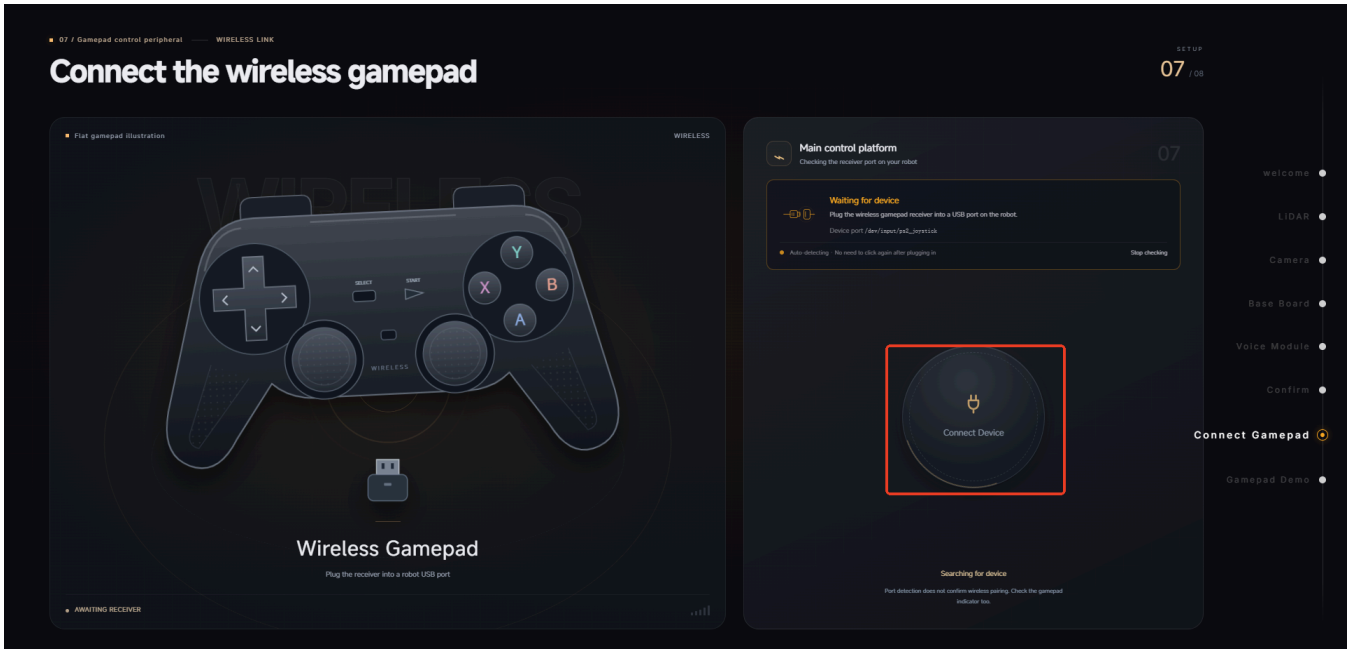




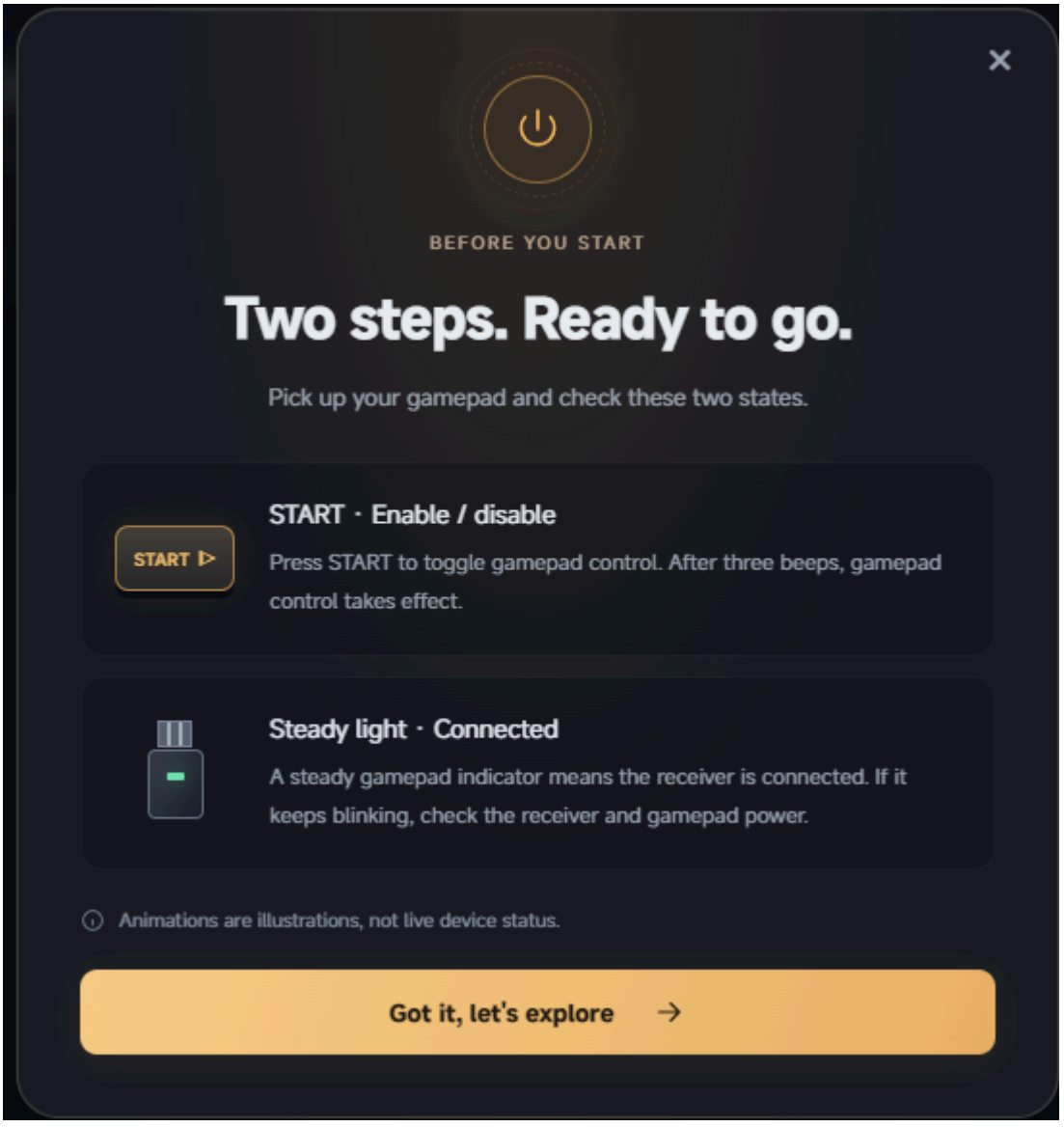
1.3 Quick Start of Joystick Control

> **Tip**
>
> For first-time use, after completing the hardware configuration, plug the joystick receiver into the USB port of the upper host controller, then click Start Joystick Control

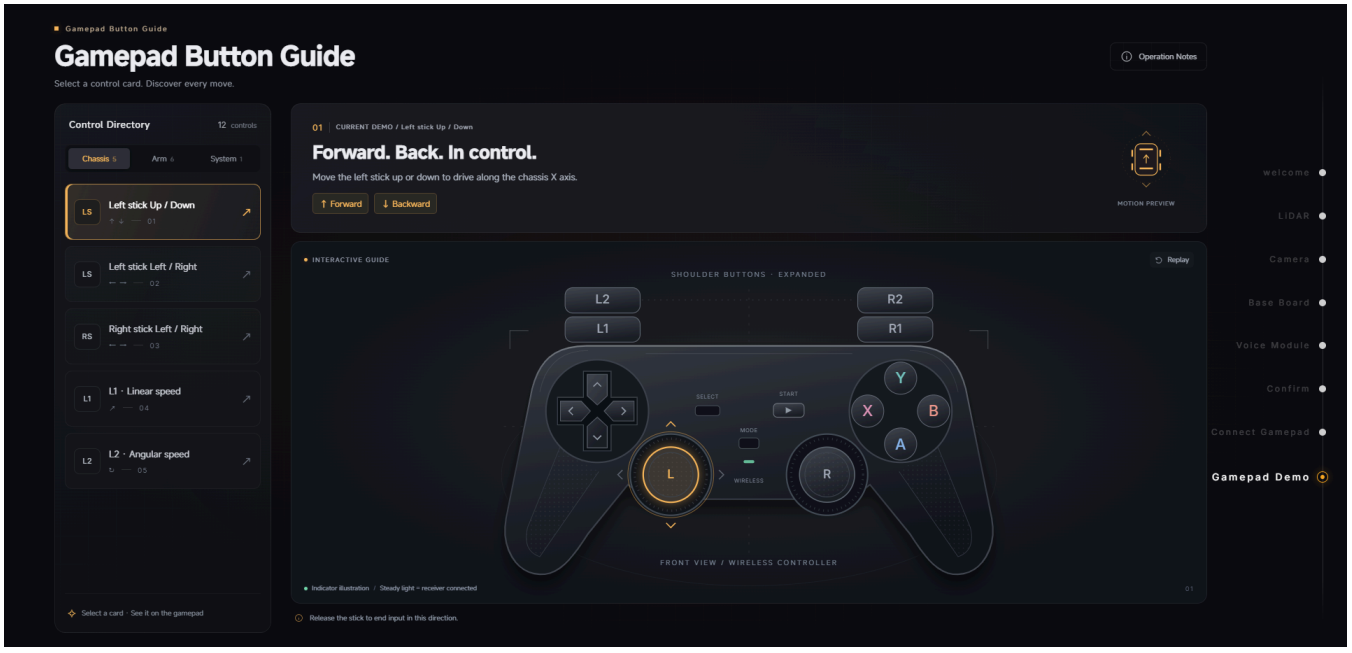
- Plug the joystick receiver into the main control board. Once the joystick receiver shows it is ready, click Start Joystick Control



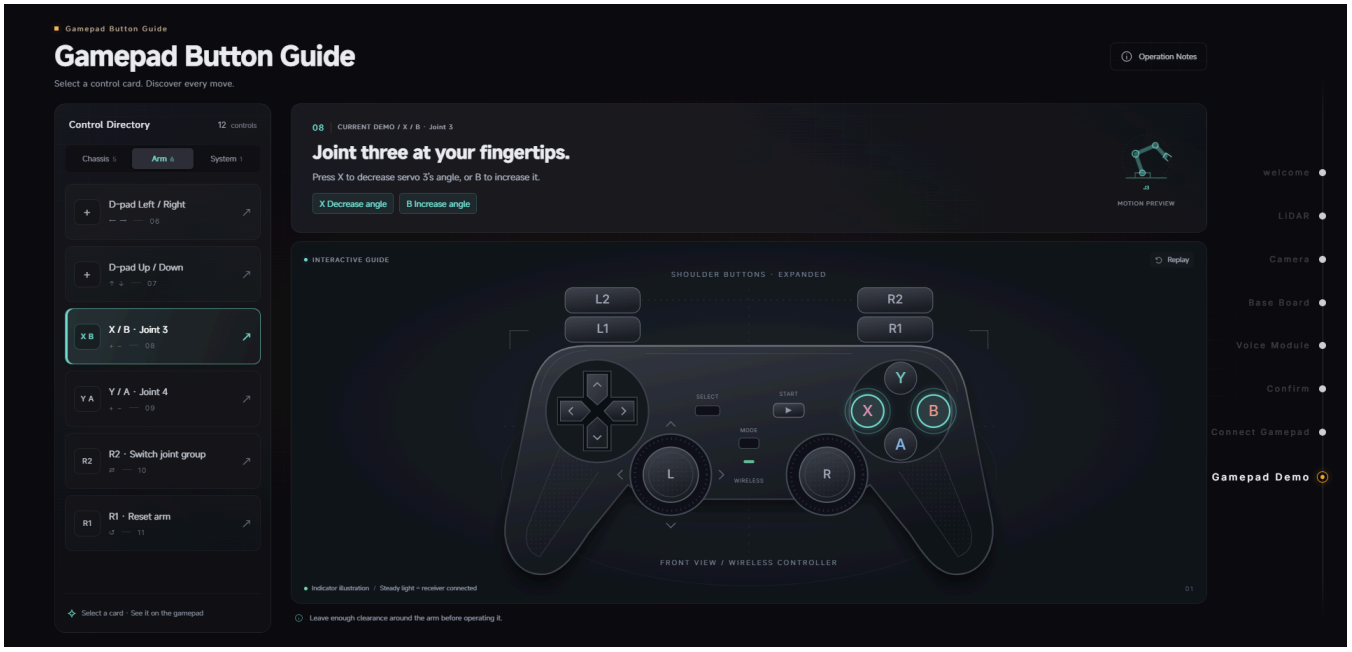
- Wait 1-2 minutes for the program to be ready. Press the START button on the joystick; after you hear a "beep-beep-beep" prompt, joystick control takes effect



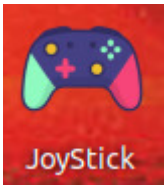
- Follow the joystick key mapping instructions on the page to quickly try out joystick control



- You can click the key cards on the left or the visualized joystick buttons to quickly view the function of the corresponding button



- In subsequent use, you can also quickly start the joystick function via the launch icon in the VNC desktop. If it needs to work together with other features such as mapping and navigation, **note that you must start the other programs first**, and start the joystick control program last



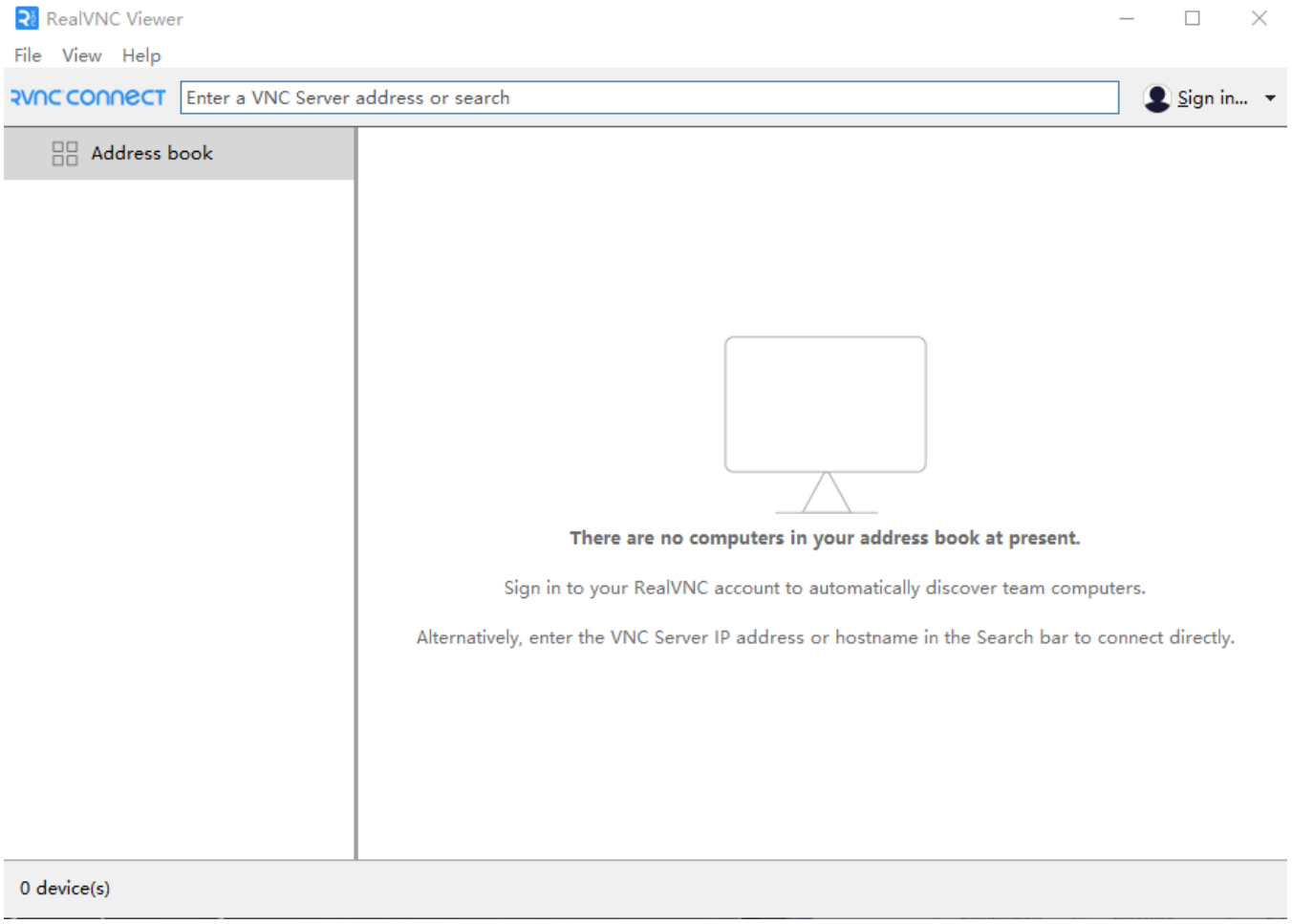
2. How to Connect to the Car

2.1 VNC Connection

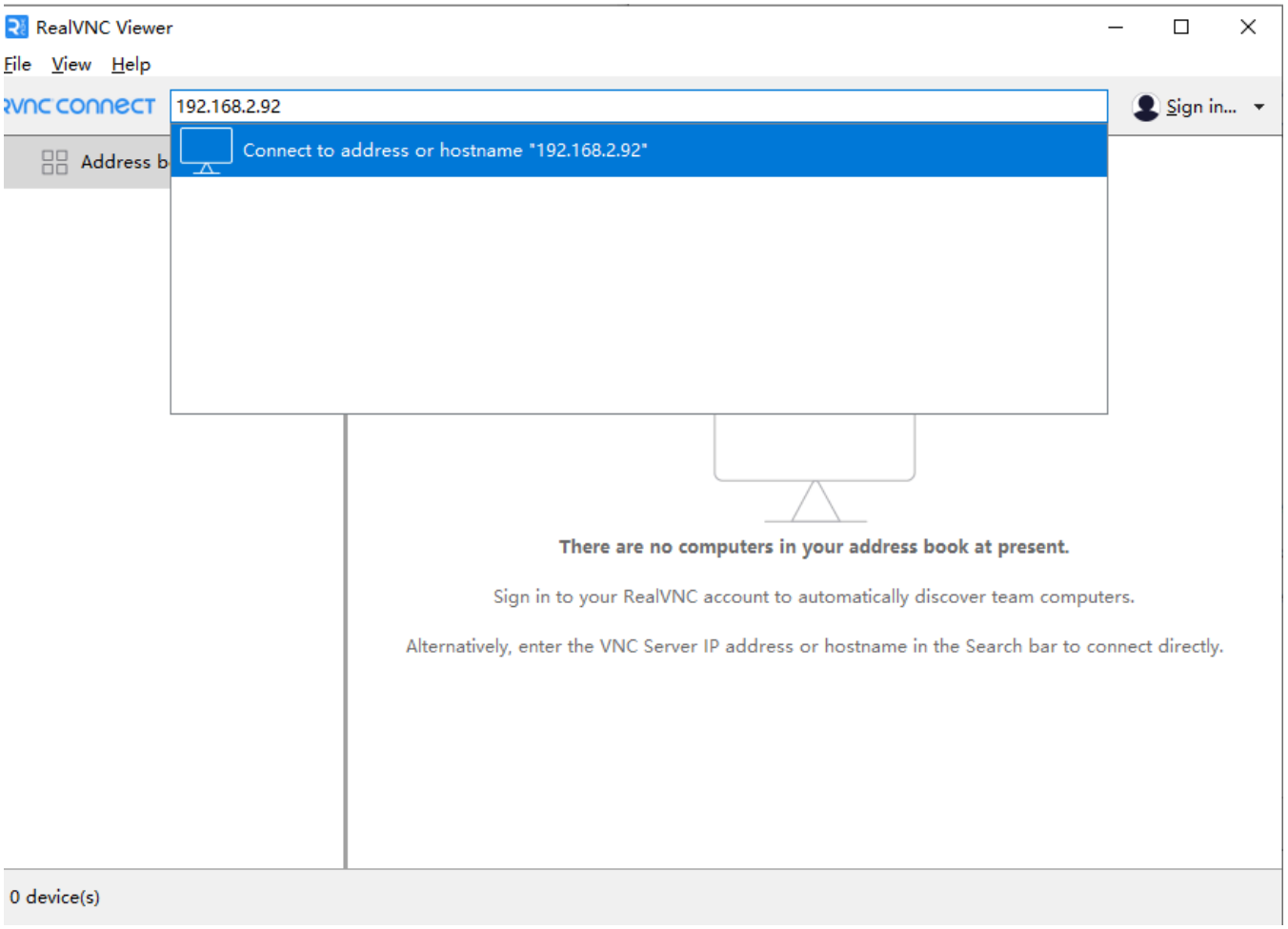
> **Tip**
>
> - There are many kinds of VNC connection tools. Here we present a recommended tool for the Windows system. Users of macOS and Linux can choose other VNC connection tools on their own; the tool is not limited to one
> - For the first connection, you can connect your computer to the car's hotspot, then log in and connect via the IP: **192.168.1.11**

- VNC allows users to remotely access and control another computer's desktop environment over the network. So when we need to access the car's desktop environment—for example, to launch a graphical display—we can use VNC to connect and log in to the car. The VNC download address is as follows: [Download VNC Viewer by RealVNC®]

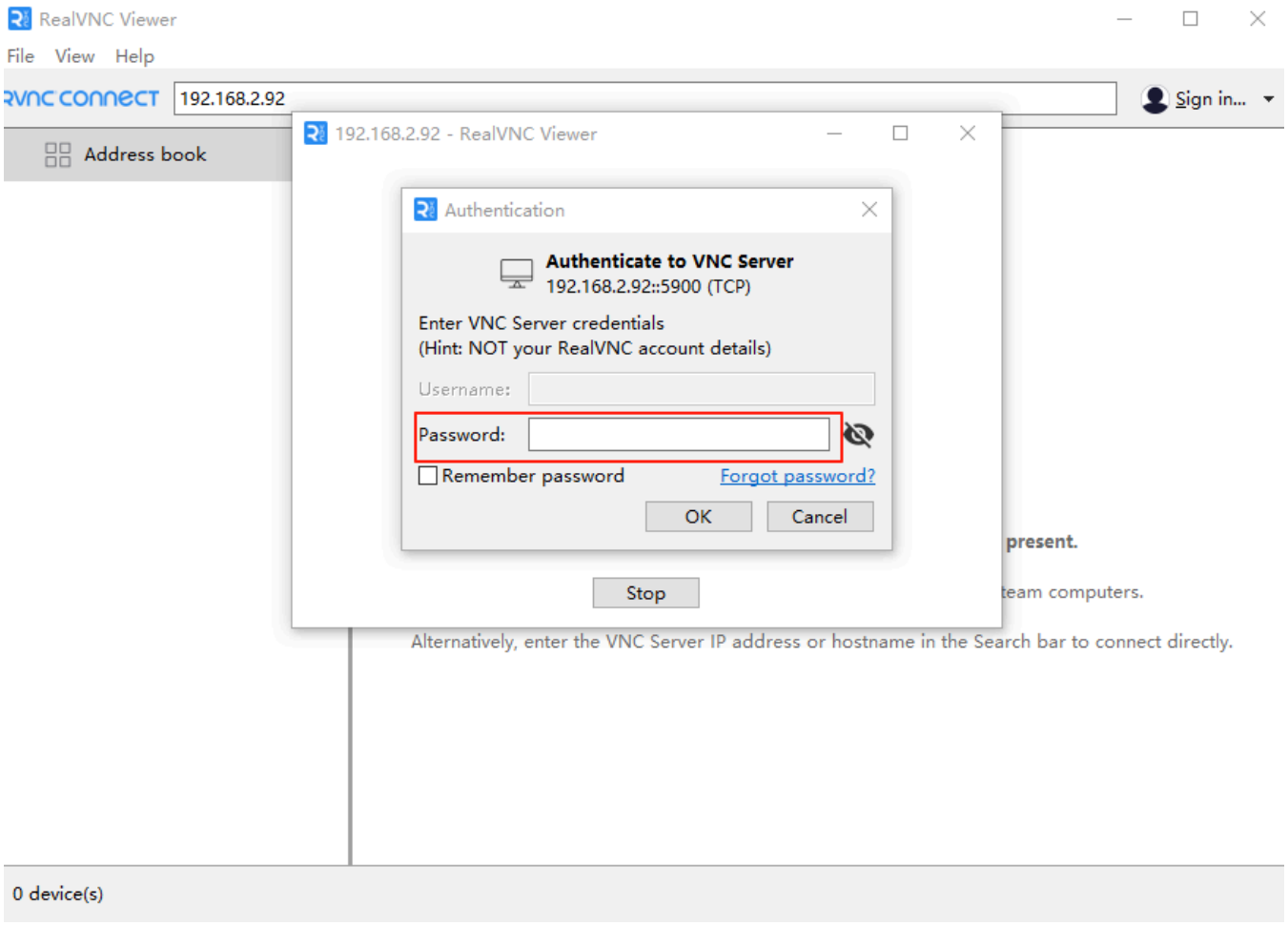
Download and install according to your computer version. After a successful installation, double-click to open it, and the screen shown is as follows



- Enter the car's IP address. Here my IP address is 192.168.2.92, as shown below,



- Then press Enter



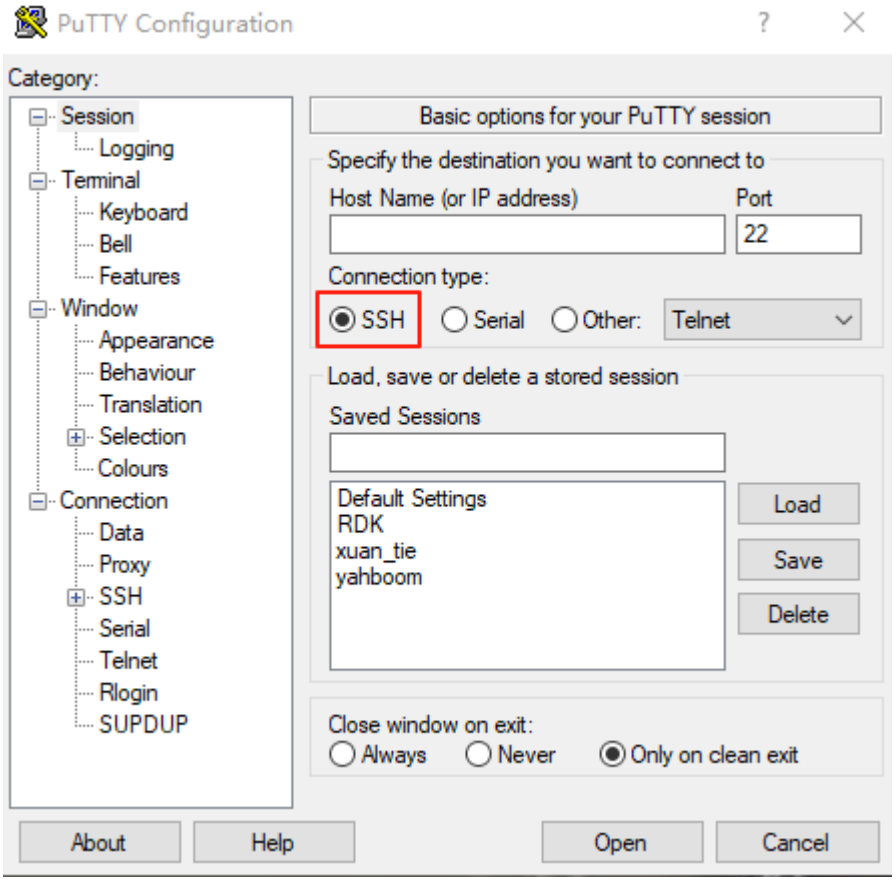
- Enter the username in Username and the password in Password. The password for all mainboards is yahboom. Then click OK to enter the desktop, as shown below



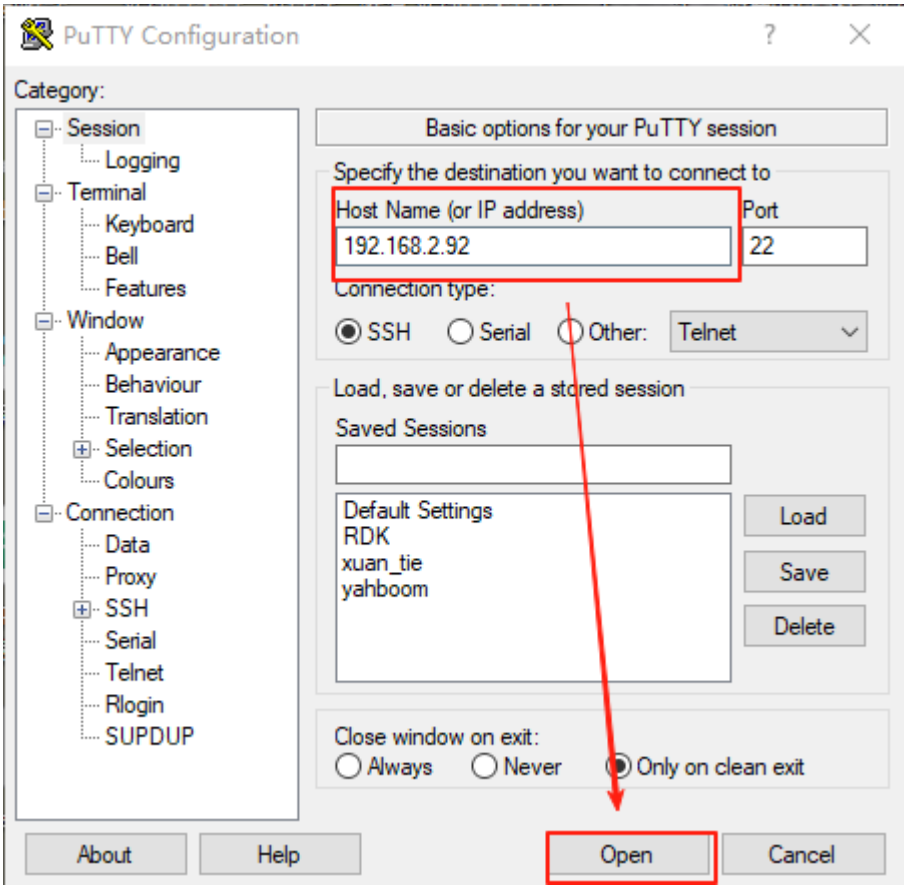
2.2 SSH Connection

> **Note**
>
> There are many kinds of SSH connection tools. Here we present a recommended SSH connection tool for the Windows system. Users of macOS and Linux can choose other SSH tools on their own

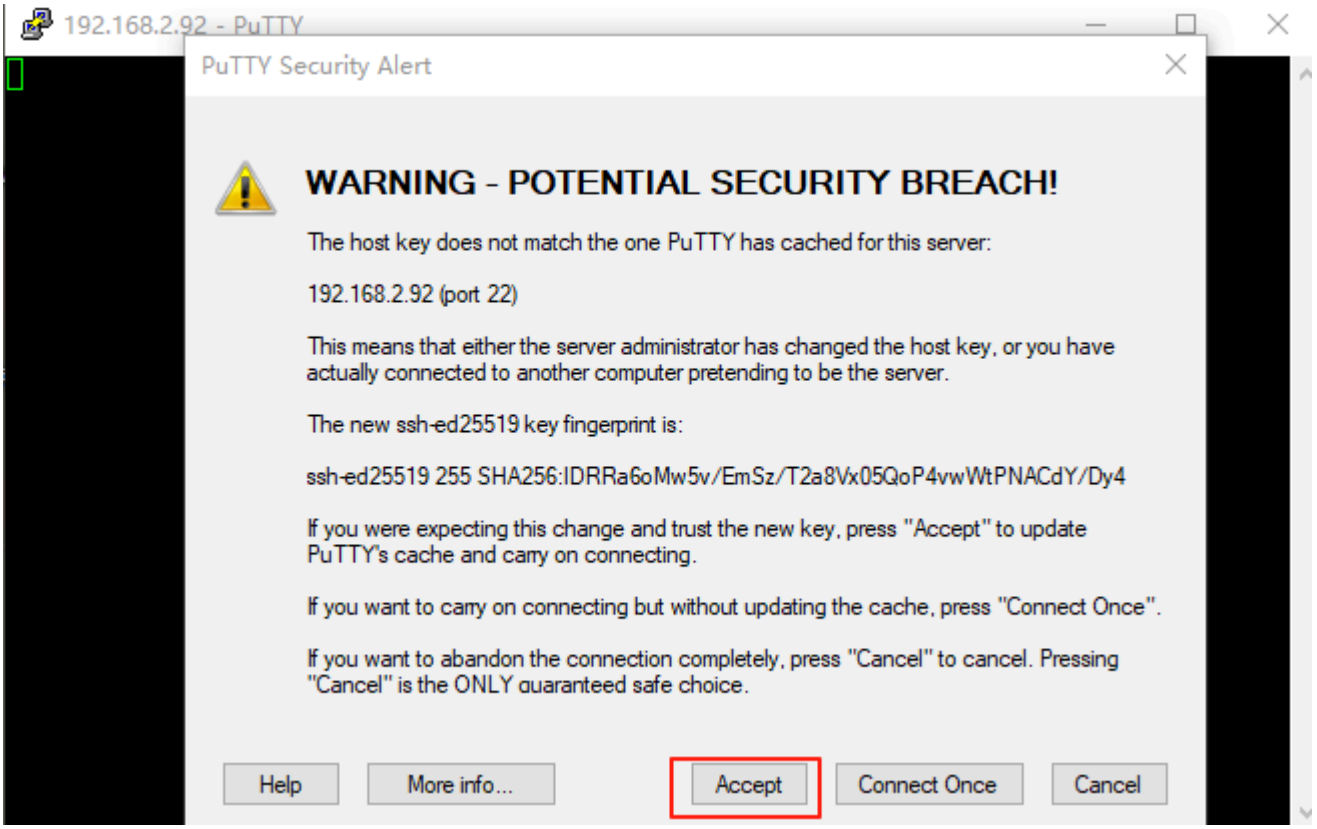
- Use PuTTY, MobaXterm, or another SSH login tool to connect to the car. Here we take PuTTY as an example. The PuTTY download address is as follows: Download PuTTY: latest release (0.83)
- Choose and install according to your computer version. After a successful installation, double-click to open it, and the PuTTY interface is as shown below



Here select **SSH**, then enter the IP address displayed on the OLED in the Host Name (or IP address) field. Here my IP address is 192.168.2.92, so enter the IP address as shown below,



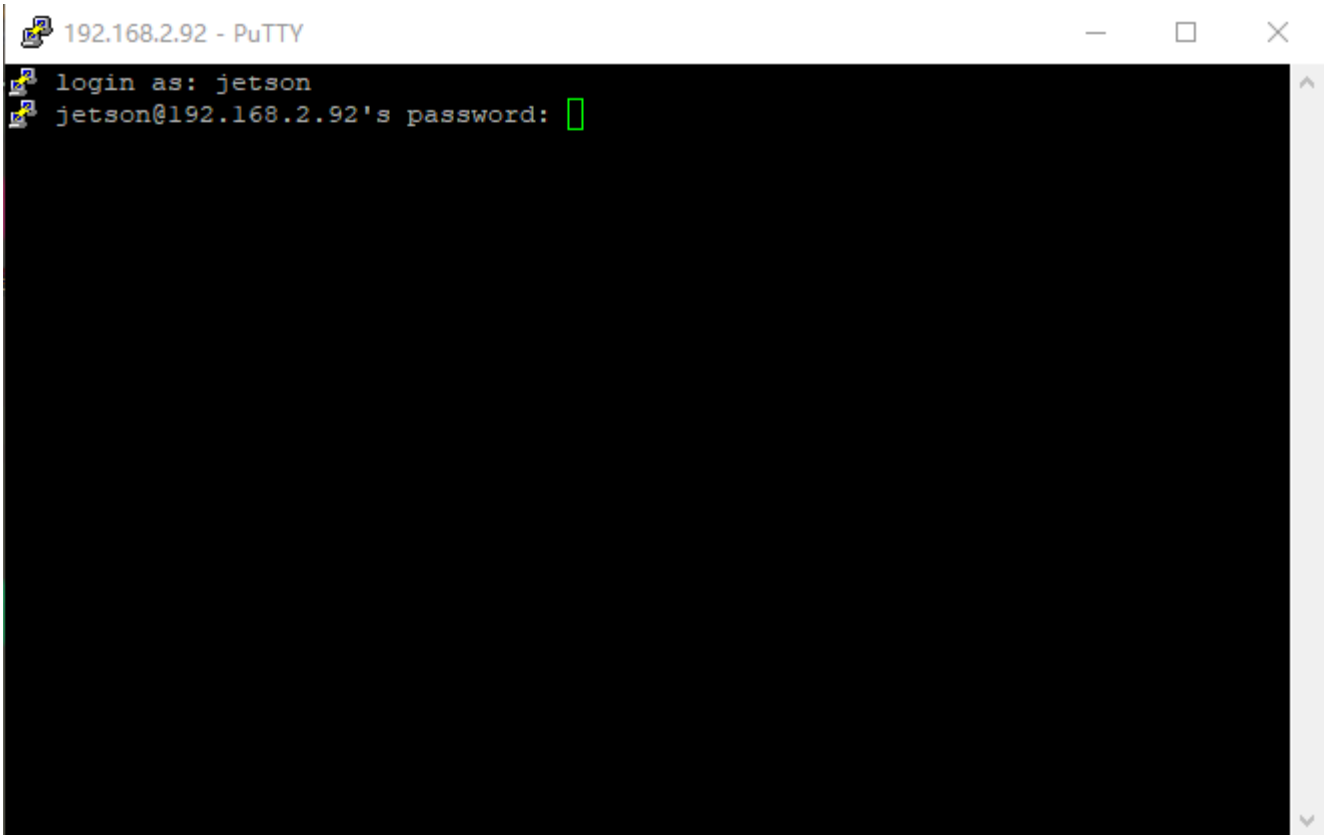
Then click Open; a terminal interface and a pop-up window will appear. Click Accept to accept, as shown below,



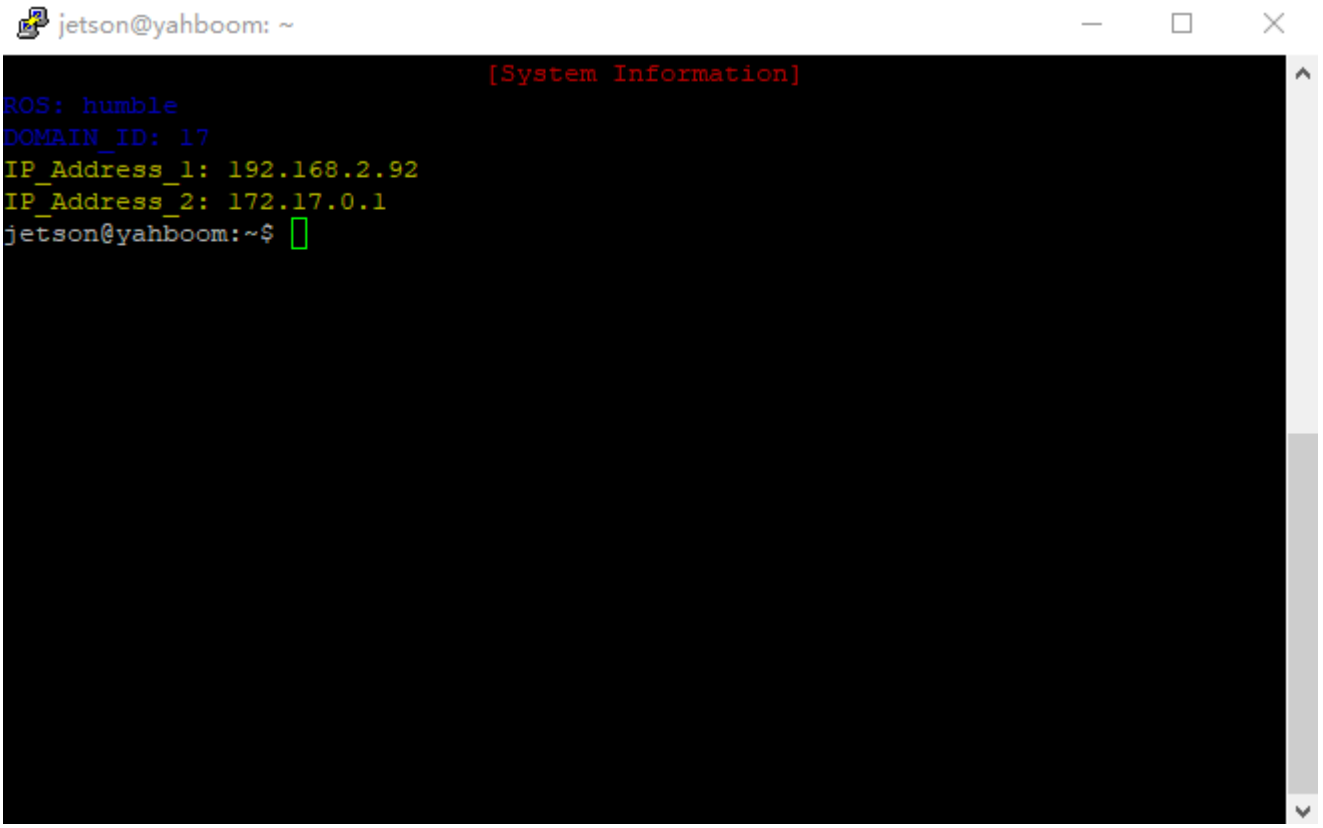
- Then the terminal displays **login as**. Enter the username for the car mainboard, then press Enter. It will then prompt for the password; enter the password. The usernames and passwords for the various mainboards are as follows:

Mainboard	Username	Password
Raspberry Pi 5	pi	yahboom
RDK X5	sunrise	yahboom
Orin-Nano	jetson	yahboom
Orin-NX	jetson	yahboom

- Here we use the Orin-Nano mainboard as an example for demonstration. Enter jetson, then press Enter, then enter the password. **Nothing is displayed while you type the password.** Enter yahboom, then press Enter



The interface after successfully connecting to the car is as follows



- After that, you can operate the car through the SSH terminal. SSH is suitable for login when a graphical interface is not needed, is far smoother than VNC, and is suitable for CLI command operations

3. Connecting the Car to Wi-Fi

> **Important**
>
> - No matter which tool you use to connect to the car, **the computer and the car must be on the same local area network**
> - The factory auto-start hotspot cannot connect to the Internet. In normal use, connect the car to Wi-Fi and operate from a computer on the same local area network

3.1 Ways to Obtain the Robot Car's IP Address

1. **Directly view the OLED screen on the car body (most intuitive)**:
   - After the car powers on, the OLED screen on the car body displays the IP address assigned to the current network interface (such as `192.168.12.57` assigned by the router or the built-in hotspot IP `192.168.1.11`).
2. **Connect to the car's built-in hotspot (offline, no router required)**:
   - On your computer, search for and connect to the car's hotspot `ROSMASTER` in the Wi-Fi list (password `12345678`).
   - After a successful connection, the car's static IP is fixed at: `192.168.1.11`.
3. **Wired Ethernet cable connection (most recommended, most stable)**:
   - Insert one end of the network cable into the car mainboard's network port and connect the other

end to the router. The car will automatically obtain a LAN IP via DHCP and display it on the OLED screen in real time.

3.2 Connecting the Robot Car to External Wi-Fi

The car's built-in hotspot has no external network access. If you later need to use AI large models, install software packages online, or update the system, it is recommended to connect the car to your home router's Wi-Fi.

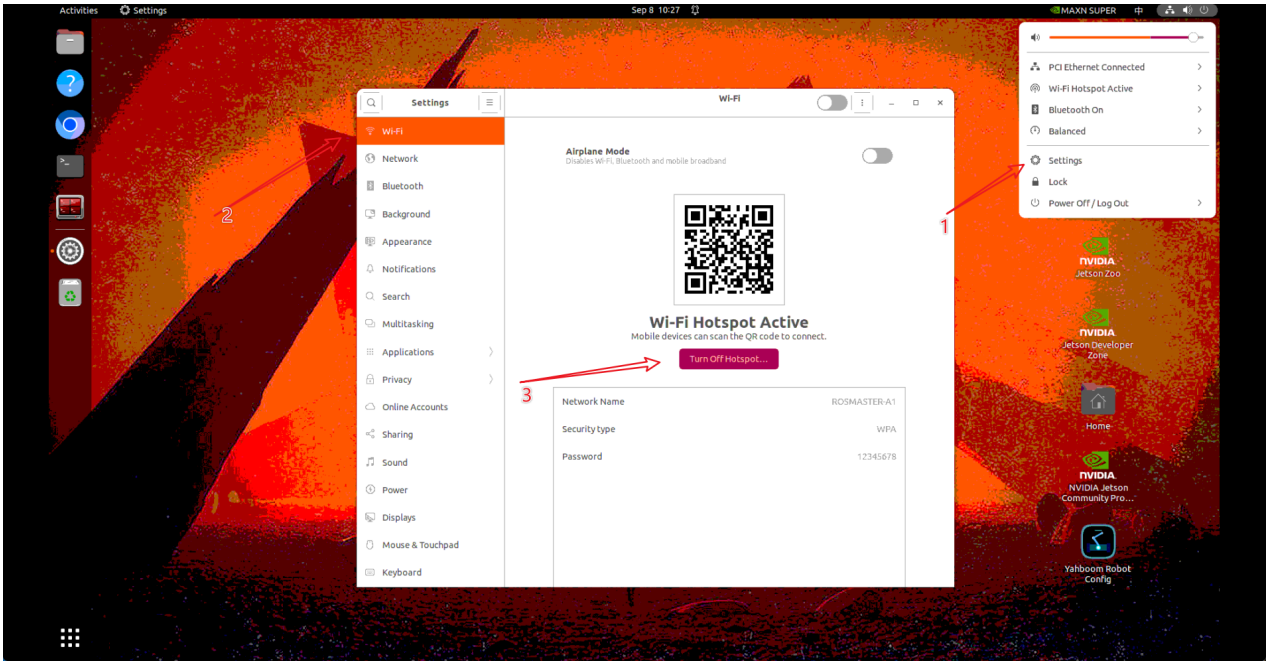
There are two common ways to connect to external Wi-Fi:

Method 1: Plug in a network cable and connect via the VNC graphical interface (recommended)

1. Insert a wired network cable into the car mainboard, read the IP address from the OLED screen on the car body (for example, `192.168.12.57`), and log in to the car using the VNC Viewer remote desktop (for detailed VNC login instructions, see Section 2.1). After a successful login, the desktop is as follows:



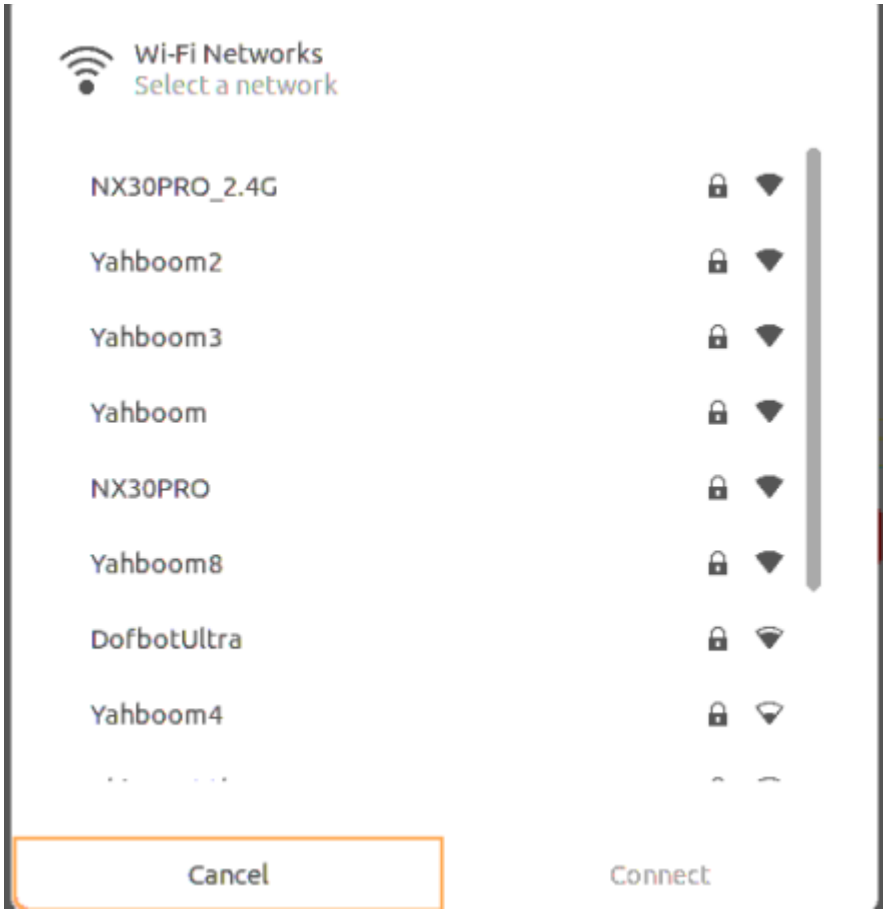
2. Open system Settings, enter the Wi-Fi settings interface, and **first turn off the car's built-in hotspot** so that the network card can scan for external Wi-Fi normally:

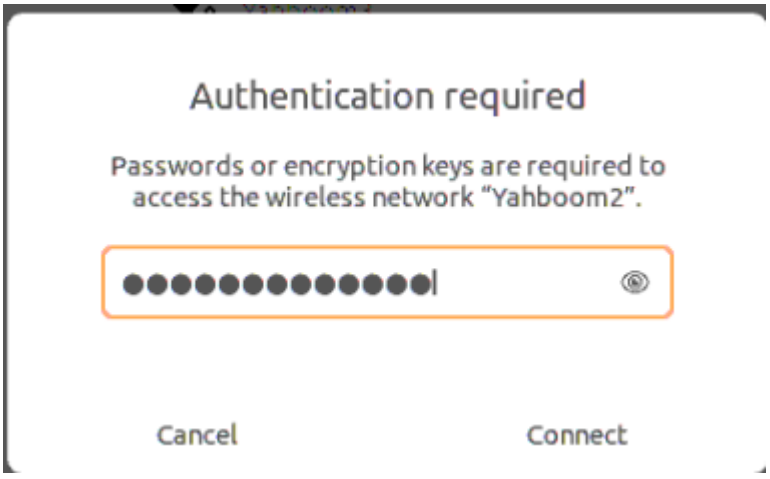


3. Click the network icon in the upper-right corner of the desktop, and click **Select Network** in the drop-down menu:

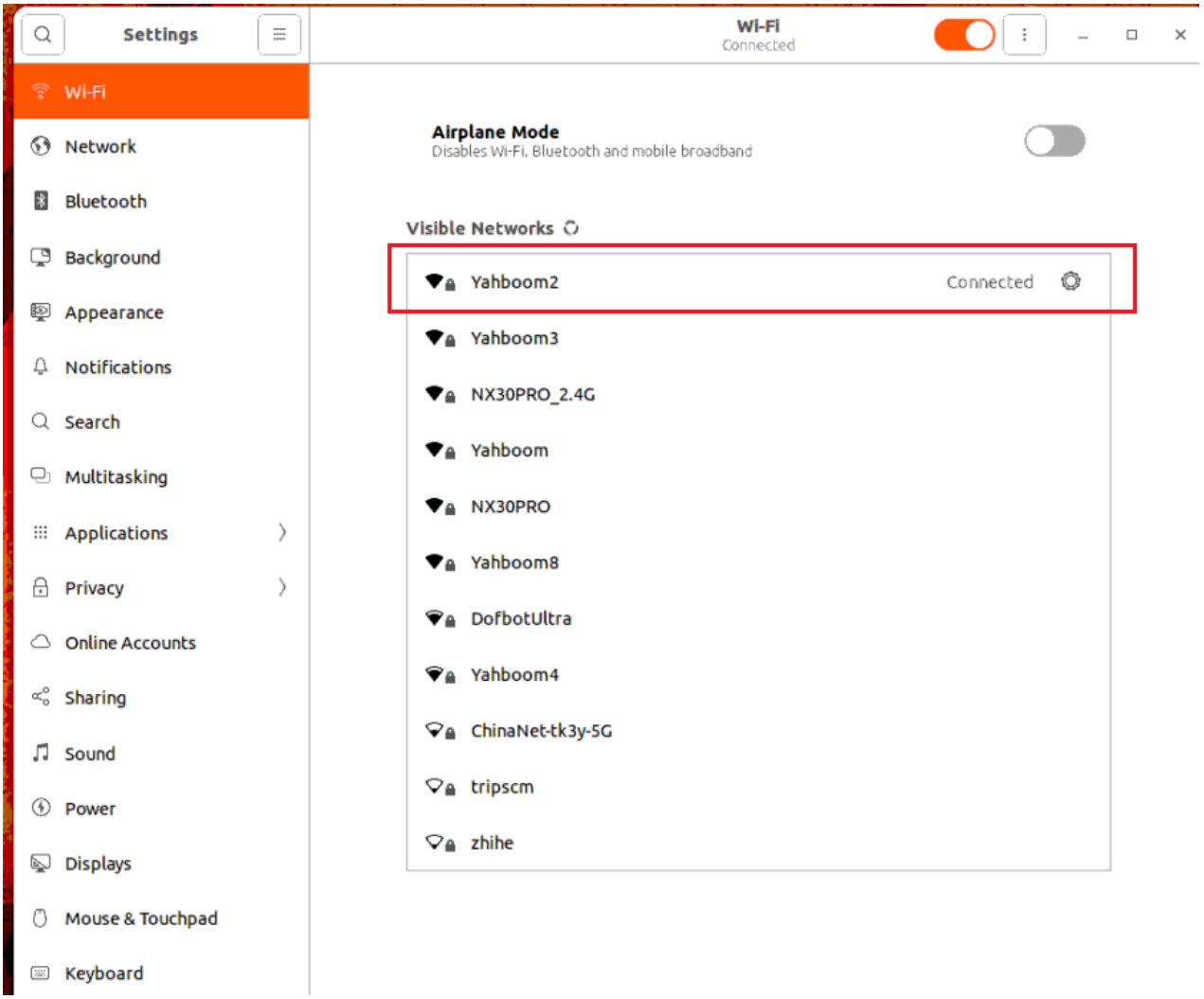


4. Select the name of the external Wi-Fi you want to connect to, enter the password, and click Connect:





5. After a successful connection, the network icon will show a connected state. At this point, unplug the network cable, and the OLED screen will also update in sync to your Wi-Fi's new IP address. You can then use this new IP for remote connections.



Method 2: When there is no network cable, configure Wi-Fi via the hotspot IP and modify the Wi-Fi priority for automatic connection

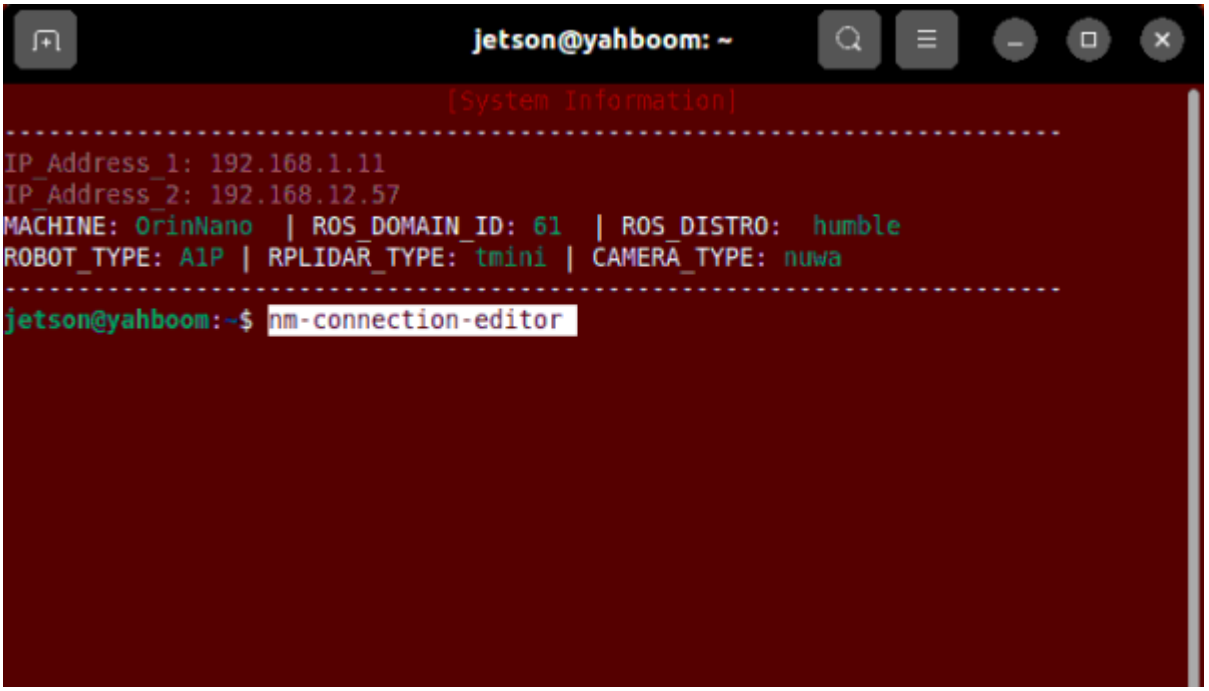
If you do not have a router network cable, you can connect your computer directly to the car's default Wi-Fi hotspot:

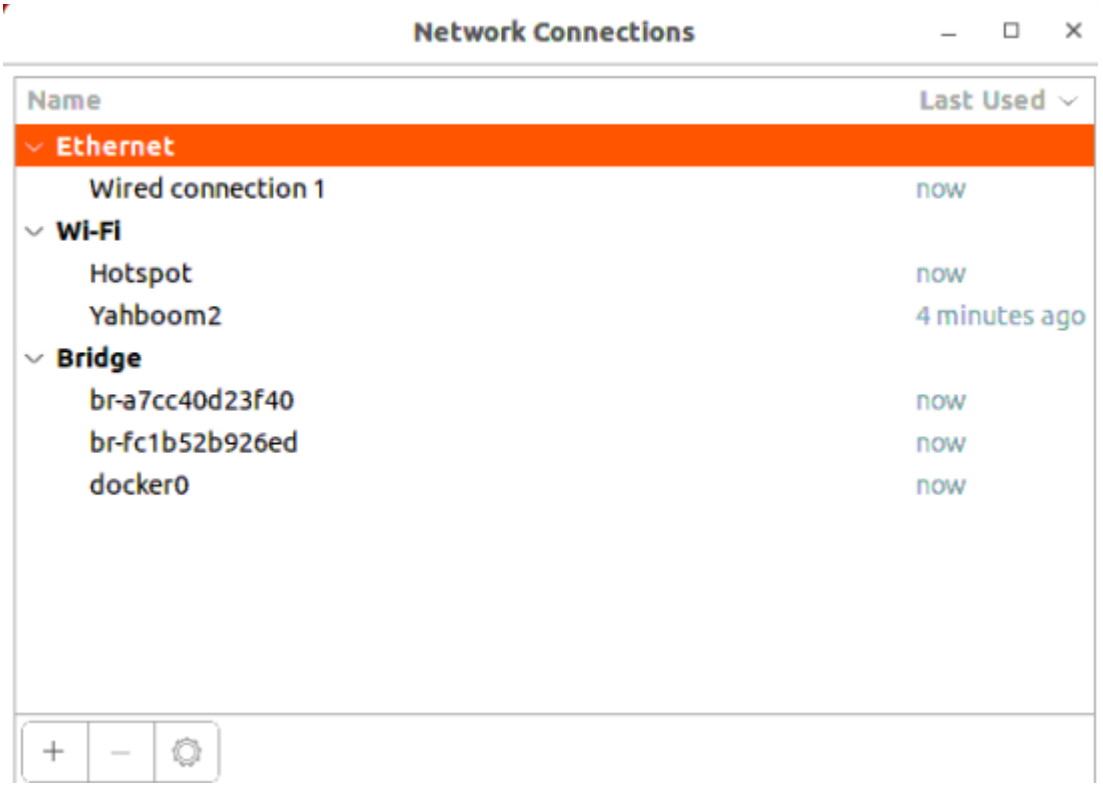
1. Connect your computer to the car's hotspot `ROSMASTER` (password `12345678`), and use VNC Viewer to connect to the hotspot's default IP: `192.168.1.11`.



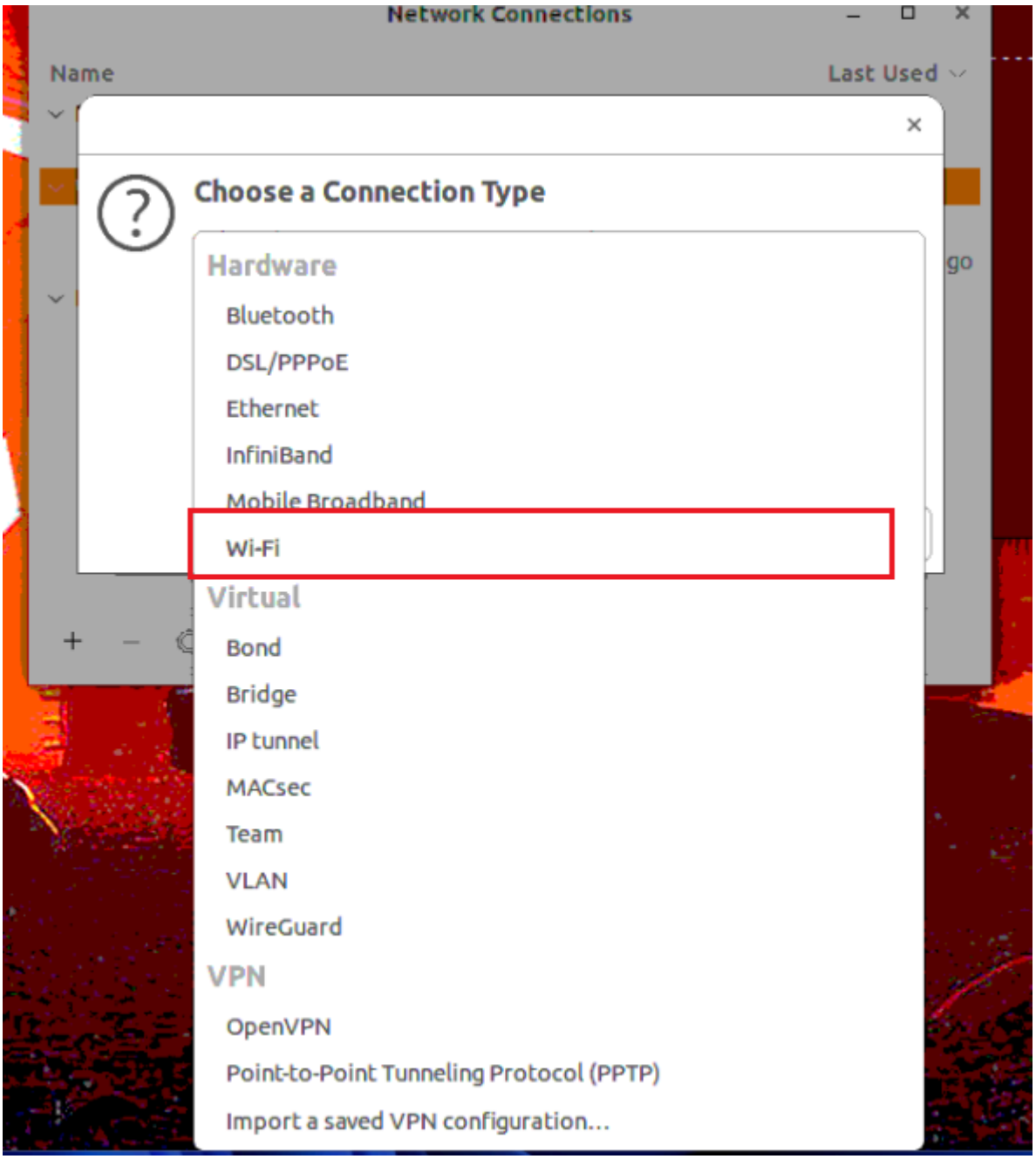
2. Open a terminal on the desktop and enter the network configuration tool command:

```
nm-connection-editor
```

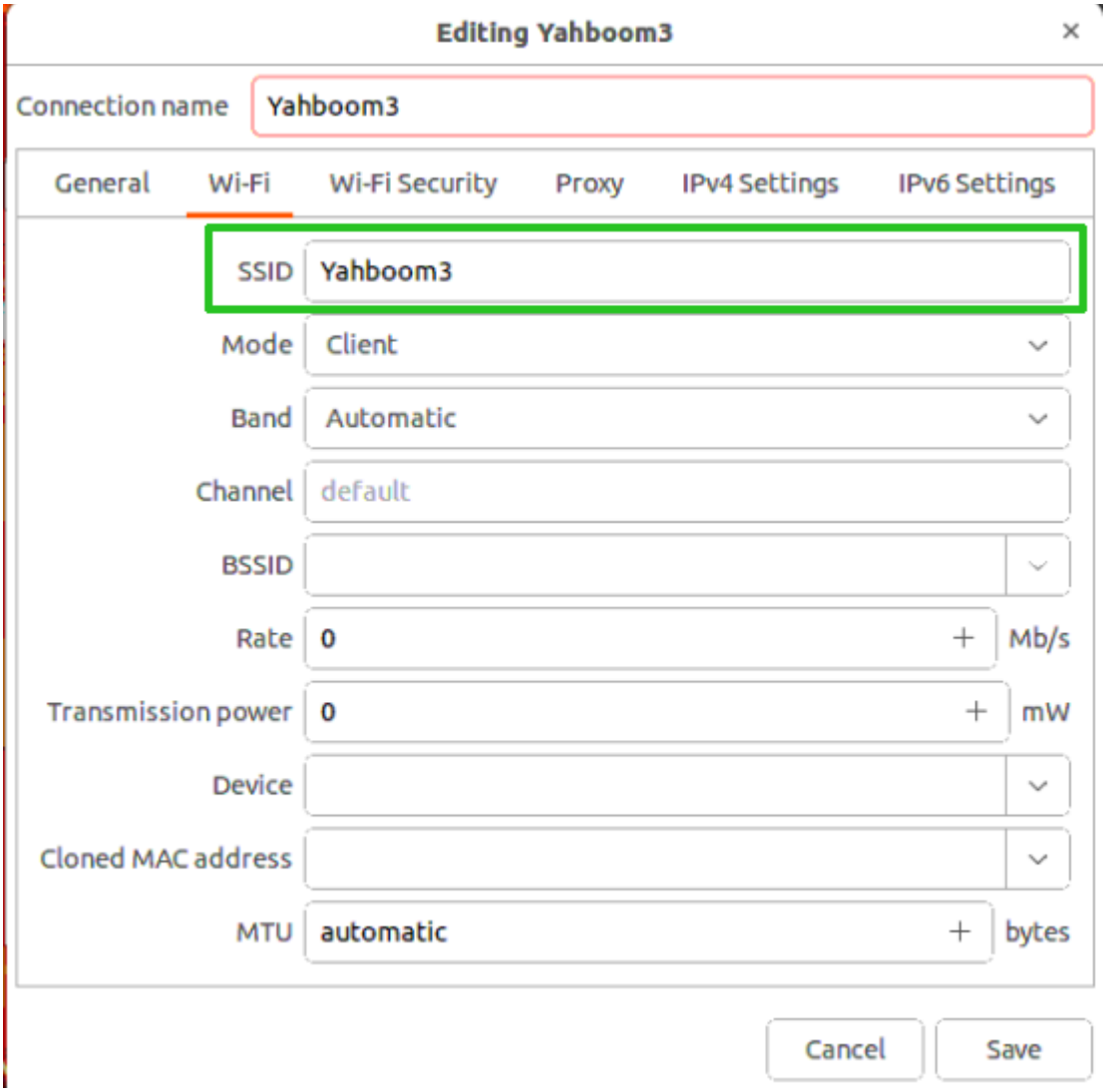




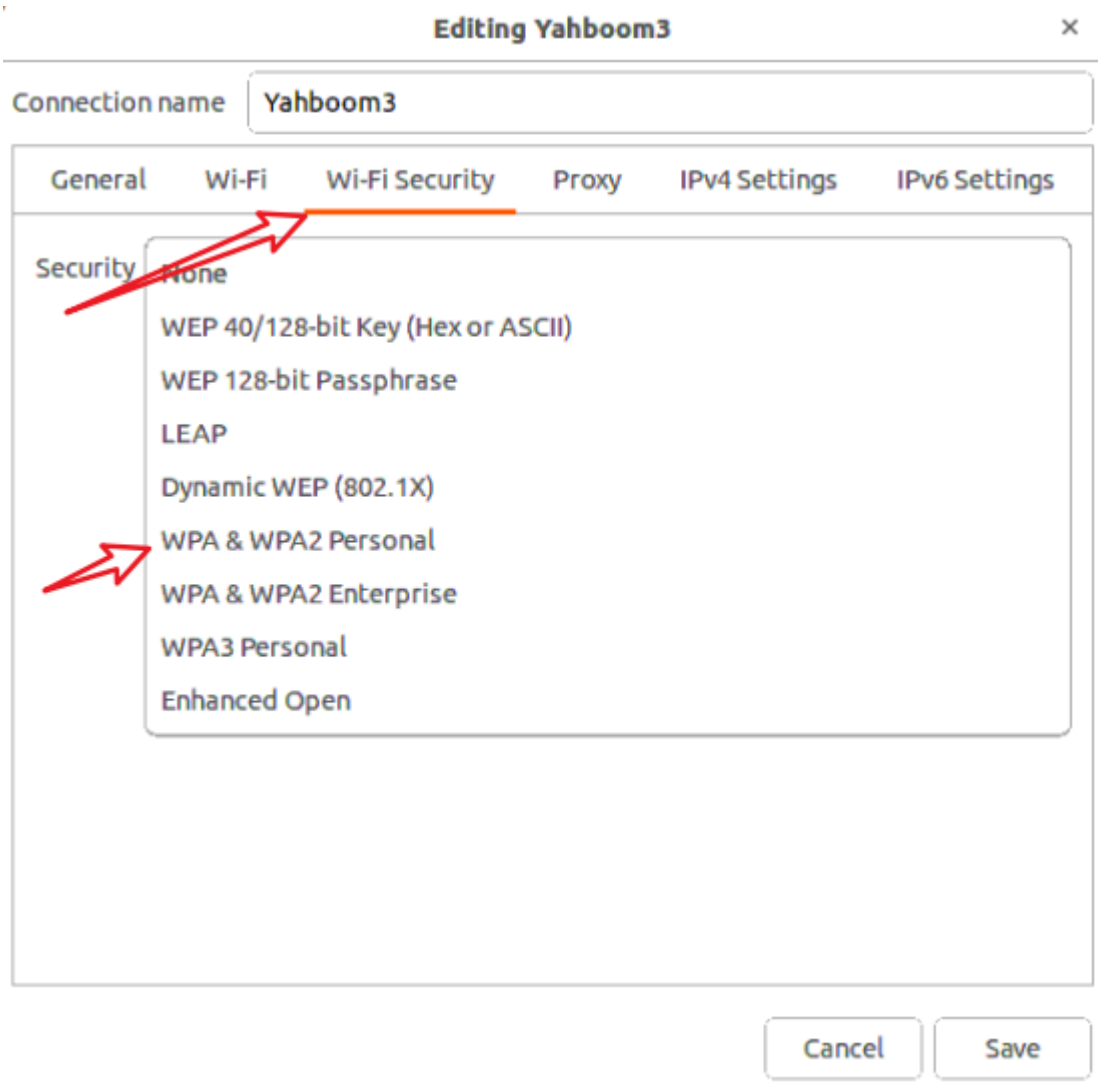
3. In the network connections window that pops up, click 【+】 in the lower-left corner to create a new connection, select **Wi-Fi** as the connection type, and click **Create**:

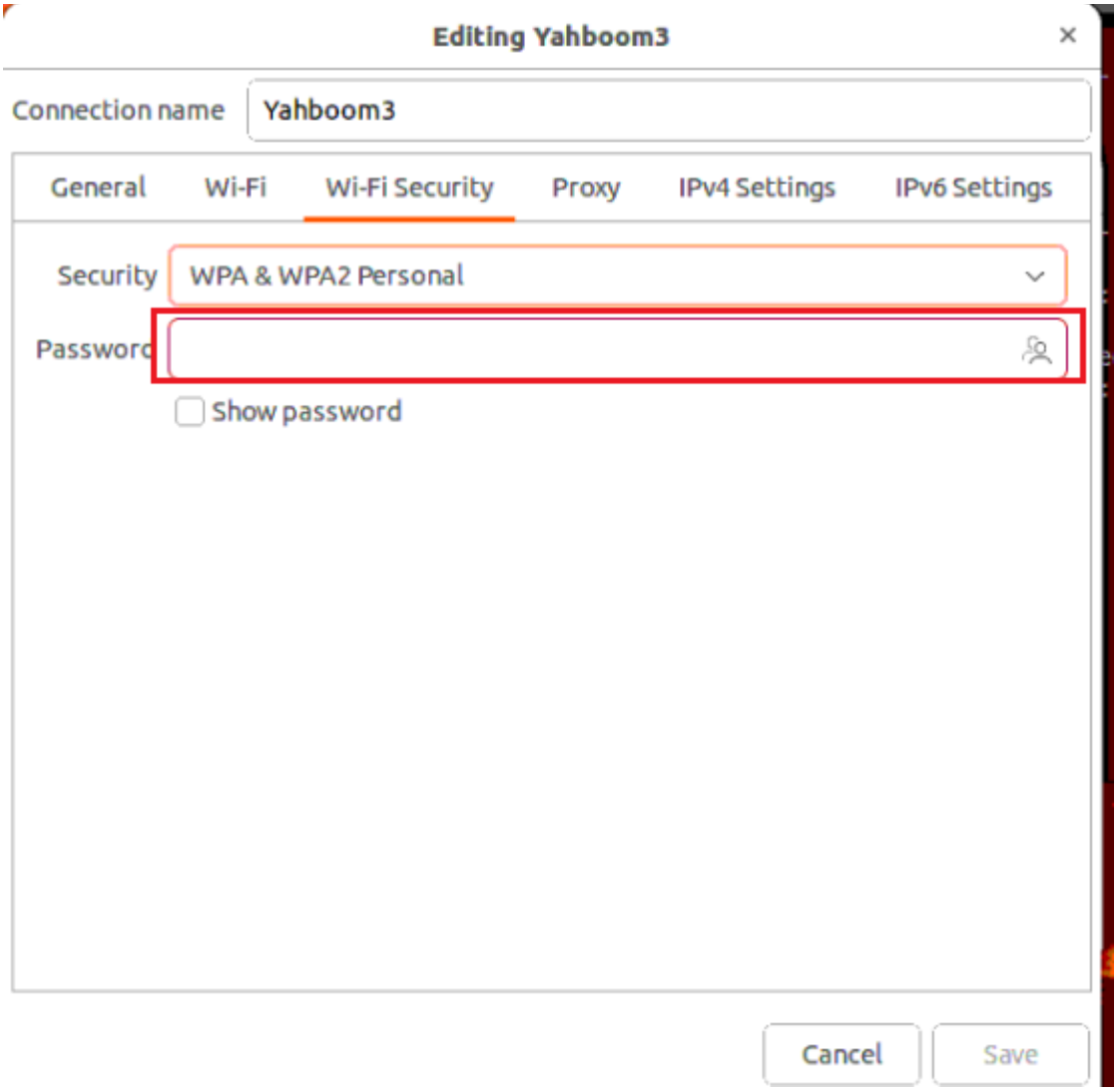


4. In the **SSID** field on the **Wi-Fi** tab, enter the name of the router Wi-Fi you want to connect to:

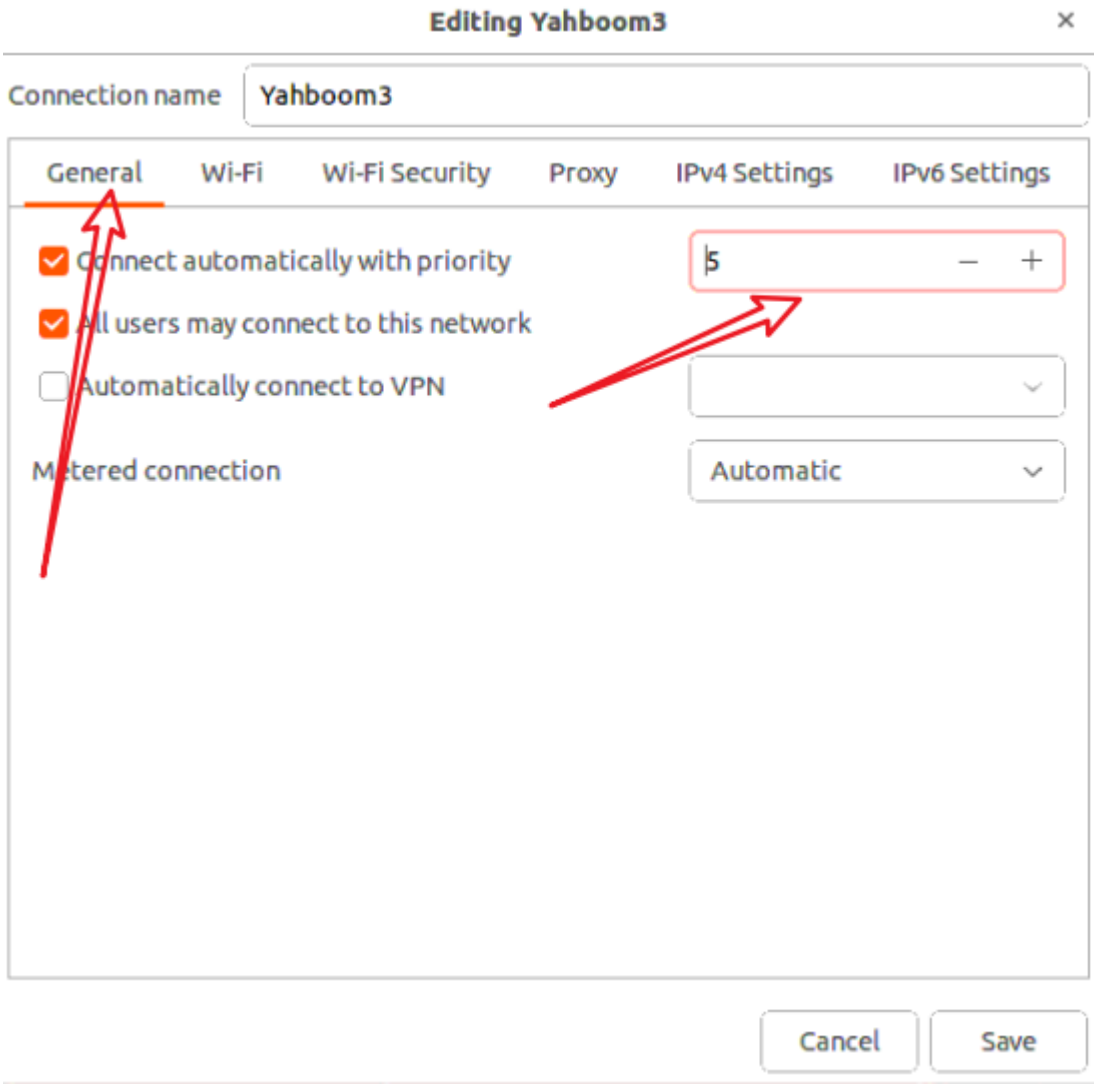


5. Switch to the **Wi-Fi Security** tab, select **WPA & WPA2 Personal** as the security type, and enter the connection password for that Wi-Fi in the **Password** field:

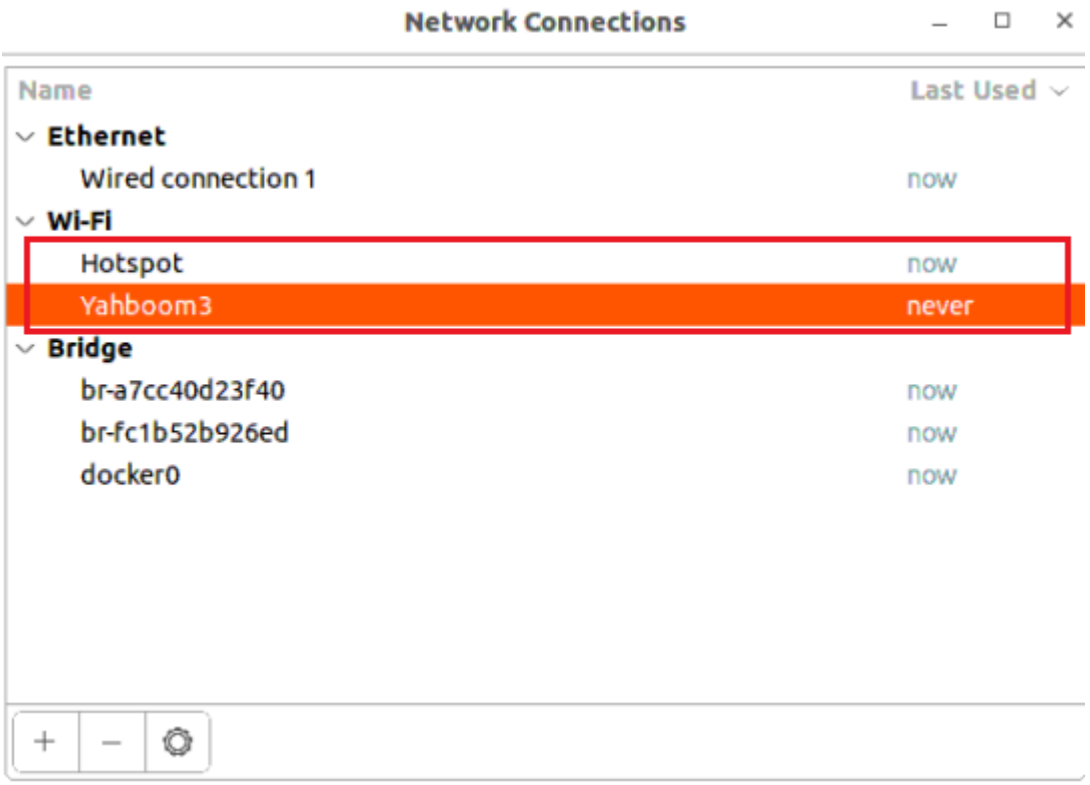

Editing Yahboom3
Connection name
Yahboom3
General
Wi-Fi
Wi-Fi Security
Proxy
IPv4 Settings
IPv6 Settings
Security
None
WEP 40/128-bit Key (Hex or ASCII)
WEP 128-bit Passphrase
LEAP
Dynamic WEP (802.1X)
WPA & WPA2 Personal
WPA & WPA2 Enterprise
WPA3 Personal
Enhanced Open
Cancel
Save



6. Switch to the **General** tab, check "Automatically connect to this network when it is available", and increase the **Priority (boot connection priority)** value (for example, set it to `10`; the larger the value, the higher the priority):



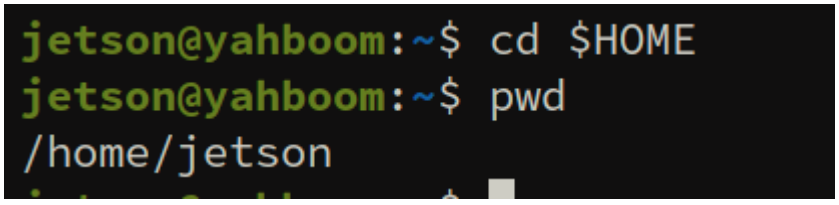
7. Click **Save** in the lower-right corner to save. Similarly, you can double-click the hotspot connection in the list to view the hotspot's priority; as long as the external Wi-Fi's priority is greater than the hotspot's priority, it is fine:



8. After the configuration is complete, close the window and restart the mainboard. After the car powers on, it will automatically prioritize connecting to the configured external Wi-Fi.

4. Tutorial Symbol Description

- The ~ and $HOME symbols that appear in the tutorial represent the mainboard's user directory in **Linux**, for example



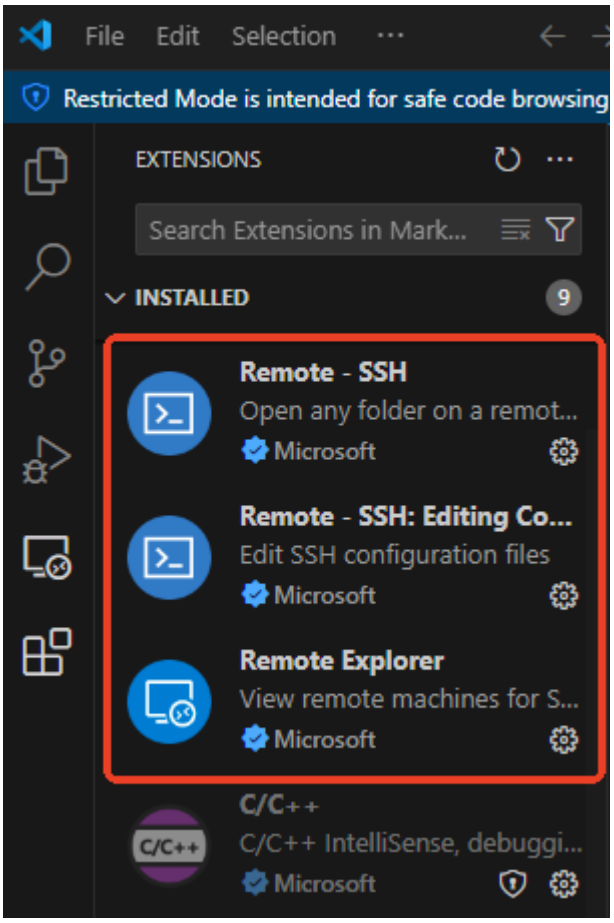
- The < xxx > symbol is a placeholder representing custom content.

5. Viewing Source Code and Compiling Packages

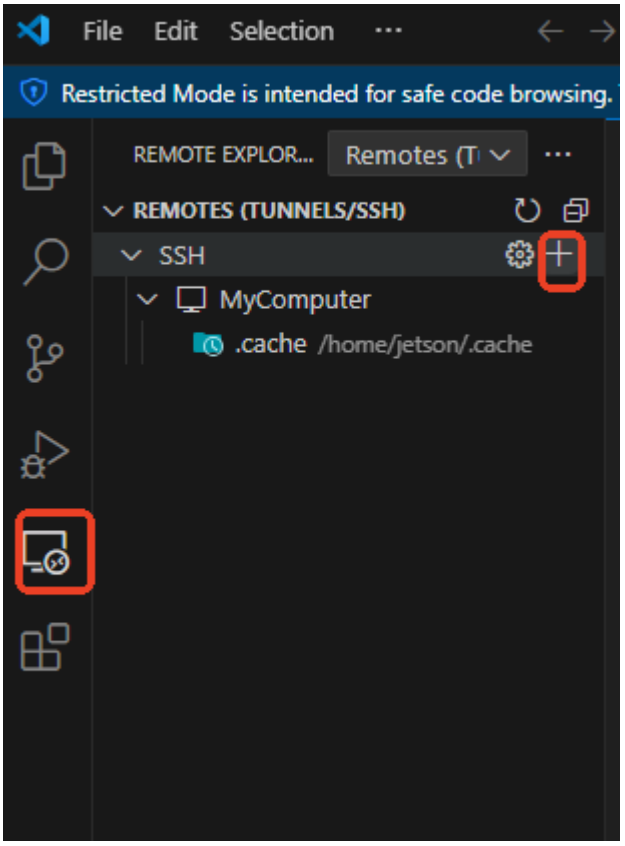
> **Important**
>
> - The Visual Studio Code here is a recommended tool. If you have your own favorite IDE, using any tool is fine.
> - Visual Studio Code is a lightweight, efficient, extensible, and integrated core development tool. Official website: https://code.visualstudio.com/

5.1 Viewing Code

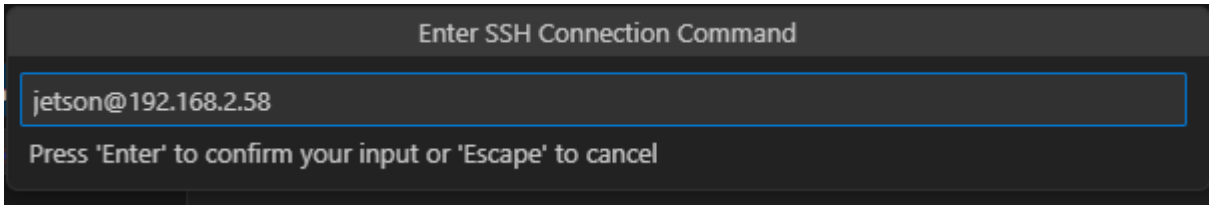
- If Visual Studio Code is not installed, you can download and install it via the link above. Launch Visual Studio Code and click the extensions on the left to install three remote connection extensions: Remote-SSH, Remote - SSH: Editing Configuration Files, and Remote Explorer



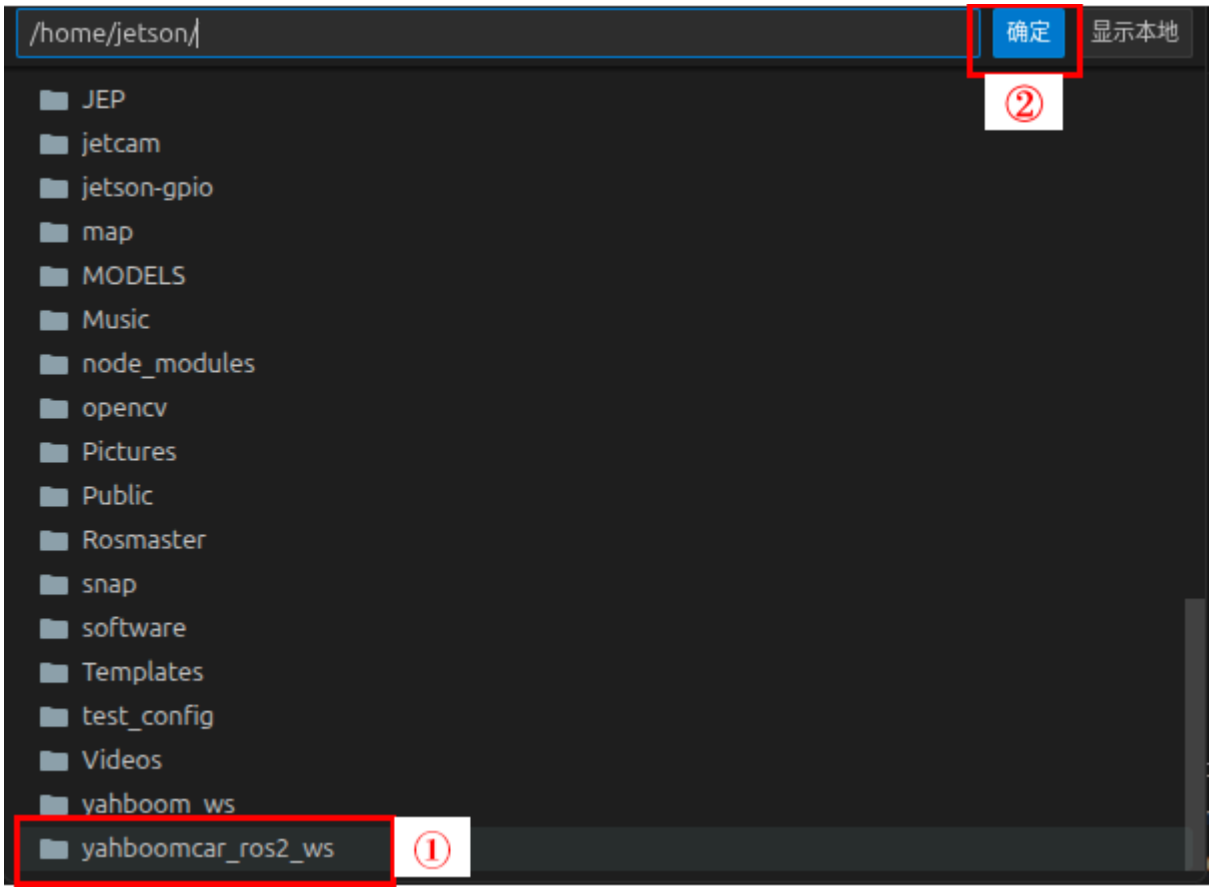
- Then click the remote connection on the left — click to create a new remote



- In the input box that pops up, enter the username@IP address according to the mainboard. After pressing Enter, enter the user password yahboom to log in to the car

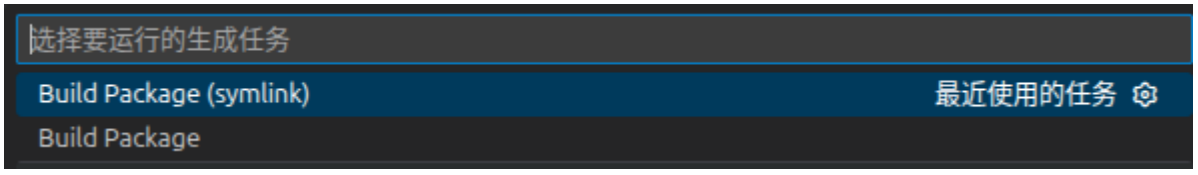


- Then click Open Folder. In the window that pops up, select the folder you want to view. Here we take yahboomcar_ros2_ws as an example, then click OK.



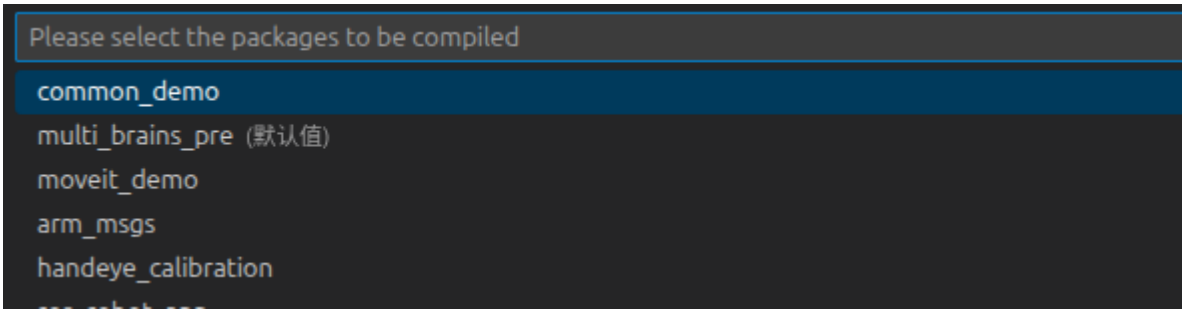
5.2 Compiling Packages

- In the VS Code interface, press the shortcut key Ctrl+Shift+B to bring up the compile options window



- Two compilation methods are provided: linked compilation and normal compilation
    - Linked compilation: yahboomcar_bringup, yahboomcar_description, m1pro_moveit_config (after modifying the package's configuration files and launch files, the changes take effect directly without recompiling)
    - Normal compilation: the source code of other packages needs to be compiled before taking effect

- After selecting the compilation option, select the name of the feature to compile, then press Enter to confirm. The command will automatically be executed to compile the corresponding package




```
* 正在执行任务: cd /home/jetson/yahboomcar_ros2_ws/yahboomcar_ws && source /opt/ros/humble/setup.bash && source install/setup.bash &&
colcon build --packages-select common_demo
```

> **Important**
>
> In yahboomcar_bringup, all configuration files and launch files take effect immediately after being modified and saved, without the need to recompile. Compilation is only needed when the user has development needs and needs to modify the source code.